📚 CCNA Study Deep Cram

📚 CCNA Study Notebook: Domain 1.0 Network Fundamentals

1.1 Explain the role and function of network components

1.1.a Routers

- **What it is:** A **Layer 3** (Network Layer) device that connects different networks and determines the best path for forwarding data packets.
- **Why it's used:** To interconnect dissimilar networks (like a home network to the internet) and to make **routing decisions** based on logical addressing (**IP addresses**).
- **Why it matters:** It is the primary device responsible for **Inter-Network Communication** (communication between different networks) and prevents broadcasts from crossing network boundaries.
- **How it works:**
  1. Receives an IP packet on one interface.
  2. Examines the **destination IP address** in the packet header.
  3. Looks up the destination network in its **routing table**.
  4. Determines the best path (next hop) to reach that network.
  5. Forwards the packet out the appropriate interface.
- **Configuration (Basic Interface Setup):**

  Cisco CLI

```
Router>enable
Router#configure terminal
Router(config)#hostname R1
R1(config)#interface GigabitEthernet0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#end
R1#show ip interface brief
```

1.1.b Layer 2 and Layer 3 Switches

Type	What it is	Function/Purpose	How it works
Layer 2 Switch	Operates at the **Data Link Layer** (Layer	Connects devices within the **same network** (LAN) and uses **MAC addresses** for forwarding decisions.	Learns MAC addresses of connected devices by inspecting incoming frames and populating its

Type	What it is	Function/Purpose	How it works
	2) of the OSI model.		**MAC Address Table**. Forwards frames out the appropriate port.
Layer 3 Switch	Operates at both **Layer 2 and Layer 3**.	Combines the functionality of a switch and a router. It can perform high-speed packet forwarding (**routing**) between VLANs/subnets while also performing L2 switching.	Uses its MAC table for L2 forwarding and its **IP Routing Table** for L3 forwarding (Inter-VLAN Routing).

- **Why they matter: L2 Switches** provide high-speed, low-latency connectivity within a LAN. **L3 Switches** enhance performance by enabling fast routing without needing a separate, dedicated router (especially for local network traffic).
- **Configuration (Basic L2 Switch Port):**

  Cisco CLI

```
Switch>enable
Switch#configure terminal
Switch(config)#interface FastEthernet0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#end
Switch#show vlan brief
```

1.1.c Next-generation firewalls (NGFW) and IPS

Term	What it is	Function/Purpose	How it works
Next-Generation Firewall (NGFW)	A stateful firewall that adds **Application Awareness**, **Intrusion Prevention**, and **Deep Packet Inspection** to traditional firewall capabilities.	Provides comprehensive security by inspecting traffic across all OSI layers, not just L3/L4. It can enforce policies based on applications (e.g., block Facebook but allow Zoom), users, and content.	Maintains state, inspects packet headers (L3/L4), and performs DPI to analyze packet payload against known signatures, behavioral models, and application definitions.
Intrusion Prevention System (IPS)	A network security appliance that monitors network traffic for malicious activity and can **proactively block** or prevent threats in real-time.	Detects and stops security exploits like buffer overflows, denial-of-service (DoS) attacks, and malware propagation based on signatures or anomaly detection.	Sits inline, analyzes traffic, and if a threat is detected, it terminates the malicious connection or drops the offending packets immediately.

- **Why they matter:** They are critical for protecting the network perimeter and internal segments from modern, sophisticated threats that traditional, port-based firewalls cannot detect.

1.1.d Access points (AP)

- **What it is:** A device that creates a Wireless Local Area Network (**WLAN**), connecting wireless clients (laptops, phones) to the wired network infrastructure.
- **Why it's used:** To provide **wireless connectivity** and mobility for end-user devices.
- **Why it matters:** It acts as the physical gateway between the wireless and wired segments of the network.
- **How it works:** Broadcasts a **Service Set Identifier (SSID)**, modulates and demodulates radio frequency (**RF**) signals to transmit and receive data, and converts wireless frames to wired Ethernet frames (and vice versa).
- **Configuration (Conceptual steps on a WLC or autonomous AP):**
  1. Define a **WLAN Profile** (SSID).
  2. Select the **radio band** (2.4 GHz or 5 GHz) and **channel**.
  3. Configure **security** (e.g., WPA3 Personal).
  4. Associate the WLAN with a **VLAN** for network segmentation.
  5. Enable the WLAN.

1.1.e Controllers (WLCs)

- **What it is:** A **Wireless LAN Controller (WLC)** is a centralized network device that manages and configures multiple Access Points (APs).
- **Why it's used:** To simplify the deployment, configuration, and management of a large number of APs, ensuring consistent security and QoS policies across the entire wireless network.
- **Why it matters:** Provides **centralized control** and automatic radio resource management, which is essential for large enterprise wireless deployments.
- **How it works:** APs operate in a **lightweight** mode, forming a secure tunnel (e.g., using **CAPWAP**) back to the WLC. The WLC handles the control plane (management) for all APs, while the data plane (traffic forwarding) can be centralized (tunneling through the WLC) or distributed (locally switching at the AP).

1.1.f Endpoints

- **What it is:** Any host device that acts as a **source or destination** of information on the network.
- **Why it's used:** To enable end-users to **access network services**, generate data, and consume information.
- **Why it matters:** They are the devices that users interact with, and they represent the most common target for security threats.

- **How it works:** Endpoints run an **Operating System (OS)**, use a **Network Interface Card (NIC)** to connect to the network, and use the **TCP/IP stack** to communicate using protocols like Ethernet and IP.
- **Examples:** Laptops, PCs, smartphones, VoIP phones, printers, IP cameras, and IoT devices.

1.1.g Servers

- **What it is:** A specialized computer or software that **provides a service** to other computers (clients) over a network.
- **Why it's used:** To centralize resources, run critical business applications, and manage essential network services.
- **Why it matters:** They host the applications and data that users and businesses depend on (e.g., websites, email, file shares).
- **How it works:** Runs a **server OS** (like Windows Server or Linux), listens on specific **ports** for client requests (e.g., port 80/443 for web, 25 for email), processes the request, and sends a response back to the client.
- **Examples:** Web Servers (HTTP/HTTPS), File Servers (SMB/NFS), Email Servers (SMTP/POP3/IMAP), DNS Servers, DHCP Servers, Authentication Servers (RADIUS/TACACS+).

1.1.h PoE (Power over Ethernet)

- **What it is:** A technology that allows network cables (like Cat 5e/6) to carry **electrical power** alongside data.
- **Why it's used:** To provide both data connectivity and power to network devices (like APs, IP phones, and IP cameras) over a single Ethernet cable.
- **Why it matters:** Simplifies installations by **eliminating the need for a separate power outlet** and corresponding electrical cabling for certain devices, allowing flexible placement.
- **How it works:** A **Power Sourcing Equipment (PSE)**—typically a PoE switch—injects power onto the unused or used data pairs of the Ethernet cable. The **Powered Device (PD)**—like an IP phone—receives the power. Modern standards (like 802.3af/at/bt) involve a negotiation process between the PSE and PD to ensure the correct amount of power is delivered.
- **Configuration (Enabling on a Cisco Switch Port):**

  Cisco CLI

```
Switch>enable
Switch#configure terminal
Switch(config)#interface FastEthernet0/24
Switch(config-if)#power inline auto  ! Default, but explicitly enables
PoE negotiation
Switch(config-if)#end
Switch#show power inline
```

1.2 Describe characteristics of network topology architectures

Network topology defines the physical or logical arrangement of connections (nodes and links) in a network.

1.2.a Two-tier

- **What it is:** A collapsed core architecture where the traditional **Core** and **Distribution** layers are combined into a single layer.
- **Why it's used:** In **smaller-to-medium-sized networks** where high availability and performance can be achieved without the complexity and cost of a dedicated core layer.
- **Why it matters:** Provides a **simpler, cost-effective** architecture with a shorter distance from the Access layer to the rest of the network.
- **How it works:** It consists of two layers:
  1. **Access Layer:** Connects end-user devices (PCs, phones, APs).
  2. **Core/Distribution Layer (Collapsed Core):** Aggregates traffic from the Access Layer, provides high-speed forwarding, and connects to the WAN/Internet.

1.2.b Three-tier

- **What it is:** A traditional, hierarchical network design model composed of three distinct layers.
- **Why it's used:** In **large, enterprise-class networks** that require high availability, scalability, high performance, and clear functional separation.
- **Why it matters:** Each layer provides a specific function, allowing for better **fault isolation**, **scalability**, and **manageability**.
- **How it works:**
  1. **Access Layer:** Connects end-user devices. Focuses on security and PoE.
  2. **Distribution Layer:** Aggregates Access Layer switches, implements policy (VLAN routing, ACLs), and provides redundancy.
  3. **Core Layer:** Provides high-speed backbone switching between Distribution Layer switches and connects to the data center/WAN. Focuses on speed and reliability.

1.2.c Spine-leaf

- **What it is:** A modern, non-blocking, two-layer data center architecture designed for high-performance and scalability.
- **Why it's used:** Primarily in **data centers** where low-latency, high-bandwidth communication is required between any two servers (often referred to as East-West traffic).
- **Why it matters:** Provides consistent, high-speed connectivity and **easy scalability**. It avoids the traditional hierarchical bottleneck.
- **How it works:**
  - **Spine Layer (Core):** High-density, high-speed switches that connect every Leaf switch.

- **Leaf Layer (Access):** Switches that connect to the servers/endpoints and are connected to *every* Spine switch.
- All Leaf switches are only one hop away from each other (via the Spine), enabling high-performance, non-blocking L3 routing (like **ECMP - Equal-Cost Multi-Path**).

1.2.d WAN (Wide Area Network)

- **What it is:** A network that connects smaller networks (LANs) across a large geographic area.
- **Why it's used:** To connect enterprise headquarters, branch offices, and remote data centers, or to connect a corporate network to the **Internet**.
- **Why it matters:** Enables **global connectivity** and access to centralized resources for geographically dispersed users.
- **How it works:** Typically utilizes services provided by a **Service Provider** (SP) like MPLS, Dedicated Leased Lines, or Internet-based VPNs. **Routers** are the primary devices at the WAN edge.

1.2.e Small office/home office (SOHO)

- **What it is:** A network designed to support a small number of users and devices, typically in a home or small business environment.
- **Why it's used:** To provide basic Internet connectivity, file sharing, and printing for a very small organization or a home.
- **Why it matters:** Characterized by **simplicity, low cost, and minimal maintenance**.
- **How it works:** Usually centers around a single **all-in-one router/modem/wireless access point** that handles routing, NAT, DHCP, and Wi-Fi access.

1.2.f On-premises and cloud

Term	What it is	Function/Purpose	Why it matters
On-premises	Infrastructure (servers, network gear, storage) that is **physically located** and maintained within the organization's own facilities.	Provides the organization with **full control** over hardware, security, and data.	Offers the highest level of control and is often preferred for legacy applications or strict regulatory environments.
Cloud	Infrastructure, platforms, or software delivered as a **service** over the Internet from a third-party provider (e.g., AWS, Azure, Google Cloud).	Allows organizations to access computing resources **on-demand** without owning the physical hardware.	Provides unparalleled **scalability, agility, and pay-as-you-go pricing**, shifting capital expenditure (CapEx) to operational expenditure (OpEx).

- **Hybrid Cloud:** The common scenario where an organization uses a combination of on-premises infrastructure and cloud services.

1.3 Compare physical interface and cabling types

1.3.a Single-mode fiber, multimode fiber, copper

Cable Type	Core Size	Light Source	Max Distance/Speed	Primary Use
Copper (Twisted Pair)	N/A	Electrical signal	Up to 100m for Ethernet (Cat 5e/6/6a)	LAN: Connecting endpoints and switches. PoE.
Multimode Fiber (MMF)	Large (50/62.5 μm)	LED/VCSEL	Shorter distances (up to 550m for 10 Gbps)	Data Center, building backbone, LAN trunks. **Shorter runs**.
Single-mode Fiber (SMF)	Small ($\sim 9 \mu m$)	Laser	Extremely long distances (10 km to over 100 km)	WAN/MAN, long-haul data, campus backbone. **Longer runs**.

- **Why they matter:** The choice of cabling depends on the required **distance**, **bandwidth**, and **cost**. Fiber is immune to electromagnetic interference (EMI) but is more costly and fragile than copper.

1.3.b Connections (Ethernet shared media and point-to-point)

Connection Type	What it is	Characteristic	Example
Ethernet Shared Media	An environment where multiple devices can access the same physical medium and must contend to transmit data.	Requires mechanisms like **Carrier Sense Multiple Access with Collision Detection (CSMA/CD)** (though mostly historical in modern switches) or the logical equivalent. Can suffer from collisions/contention.	Older Hubs, Wireless networks. Modern switches break this into a **single collision domain per port**.
Point-to-Point	A direct, dedicated link between **exactly two** communication endpoints.	Provides a **dedicated medium** with full bandwidth between the two devices, eliminating the possibility of collisions.	A trunk link between two switches, a WAN link between two routers, or a single PC connected to a switch port.

1.4 Identify interface and cable issues (collisions, errors, mismatch duplex, and/or speed)

Issue	Description	Symptom/Indicator	Troubleshooting Command
Collisions	Two devices on a shared segment (like a hub or half-duplex link) attempt to transmit at the same time.	High **late collision** or **collision** counter on the interface statistics. Slow network performance.	`show interfaces [interface-id]`
Errors	Frames/packets received with issues that prevent processing (e.g., CRC errors, runts, giants).	High **input errors** or **CRC** counters. Data corruption, retransmissions.	`show interfaces [interface-id]`
Duplex Mismatch	One side of a link is configured for **full-duplex** (transmitting and receiving simultaneously) while the other is in **half-duplex** (one at a time).	High **input errors**, **collisions**, and very poor throughput. The half-duplex side incorrectly assumes the wire is clear.	`show interfaces [interface-id]`: Check the *Duplex* field.
Speed Mismatch	The configured or negotiated speed (e.g., 100 Mbps vs. 1 Gbps) is different on each side of a link.	**No link establishment** or link flapping (often due to misconfiguration rather than simple mismatch, as auto-negotiation usually handles this). Check the *Speed* field.	`show interfaces [interface-id]`

- **Action:** For duplex and speed mismatches, ensure both sides of the link are configured for the **same setting**, preferably **auto-negotiation** unless a specific requirement dictates manual configuration.

1.5 Compare TCP to UDP

The Transport Layer (Layer 4) uses these two primary protocols:

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection Type	**Connection-Oriented** (Requires a **three-way handshake** before data transfer).	**Connectionless** (No handshake required).
Reliability	**Reliable**. Ensures delivery via acknowledgments (ACKs) and retransmissions.	**Unreliable** (Best-effort). No retransmissions or ACKs.
Flow Control	Yes (Prevents the sender from overwhelming the receiver).	No.
Error Checking	Yes (Checksum).	Yes (Checksum), but no error recovery.

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Header Overhead	**Higher** (20 bytes minimum). Includes sequence and acknowledgment numbers.	**Lower** (8 bytes minimum).
Speed	**Slower** (due to overhead and reliability checks).	**Faster** (low overhead).
Uses/Examples	Web (HTTP/HTTPS), File Transfer (FTP/SFTP), Email (SMTP, IMAP, POP3), SSH, Telnet.	DNS queries, DHCP, VoIP/Streaming (RTP), Video Conferencing, SNMP.

1.6 Configure and verify IPv4 addressing and subnetting

IPv4 Addressing

- **What it is:** A 32-bit numerical address used to uniquely identify a device on an IPv4 network. It's written in dotted-decimal notation (e.g., 192.168.1.1).
- **Purpose:** Enables **logical addressing** at Layer 3 (Network Layer) for end-to-end communication.
- **Components:** Every IP address has two parts: the **Network ID** and the **Host ID**. These are separated by the **Subnet Mask**.

Subnetting

- **What it is:** The process of dividing a large IP network into smaller, more manageable sub-networks (subnets).
- **Why it's used:**
  1. **Efficiency:** Reduces the size of the broadcast domain, which conserves bandwidth and improves performance.
  2. **Organization:** Allows for better structure and allocation of IP addresses.
  3. **Security:** Permits the implementation of security policies between subnets.
- **How it works:** By borrowing bits from the Host ID portion of the IP address and adding them to the Network ID using a more restrictive subnet mask.
  - **Example (CIDR Notation):** A **/24** mask (255.255.255.0) means 24 bits are used for the Network ID and 8 bits for the Host ID. A **/26** mask (255.255.255.192) means 26 bits are used for the Network ID and 6 bits for the Host ID, creating four smaller subnets.

Configuration (Cisco Router/Switch):

Cisco CLI

```
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
```

Verification:

Cisco CLI

```
Router#show ip interface brief  ! Verifies the IP and status
Router#ping 192.168.10.2        ! Verifies connectivity to another host
```

1.7 Describe private IPv4 addressing

- **What it is:** IP address ranges reserved for use within a **private network** (LAN) and are **not routable** on the public Internet.
- **Why it's used:** To provide a sufficient supply of internal IP addresses without depleting the limited public IPv4 address space.
- **Why it matters:** Devices using private addresses require **Network Address Translation (NAT)** to communicate with the public Internet.
- **Private IP Address Ranges (Defined by RFC 1918):**

Class	Network Range	Subnet Mask (Default)	CIDR Notation
Class A	10.0.0.0 to 10.255.255.255	255.0.0.0	/8
Class B	172.16.0.0 to 172.31.255.255	255.255.0.0	/12
Class C	192.168.0.0 to 192.168.255.255	255.255.255.0	/16

1.8 Configure and verify IPv6 addressing and prefix

IPv6 Addressing

- **What it is:** A 128-bit numerical address written in hexadecimal notation (e.g., `2001:db8:acad:1::1`).
- **Purpose:** To replace IPv4, providing a massive address space to accommodate the exponential growth of the Internet.
- **Prefix:** In IPv6, the subnet mask is replaced by the **prefix length**, written after a slash (e.g., `/64`). A `/64` prefix is the standard for a single IPv6 subnet, using the first 64 bits for the Network Prefix and the last 64 bits for the Interface ID (Host part).

Configuration (Cisco Router):

Cisco CLI

```
Router(config)#ipv6 unicast-routing ! Must be enabled globally
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ipv6 address 2001:db8:acad:1::1/64
Router(config-if)#no shutdown
```

Verification:

Cisco CLI

```
Router#show ipv6 interface brief  ! Verifies the IPv6 address and status
Router#ping 2001:db8:acad:1::2  ! Verifies connectivity to another host
```

1.9 Describe IPv6 address types

1.9.a Unicast

An address that uniquely identifies a single interface. A packet sent to a unicast address is delivered to the single interface identified by that address.

- **Global Unicast Address (GUA):**
  - **What it is:** Globally routable and unique address, analogous to public IPv4 addresses.
  - **Format:** Typically starts with `2000::/3`.
  - **Purpose:** Allows a device to be reachable from anywhere on the public IPv6 Internet.
- **Unique Local Address (ULA):**
  - **What it is:** Routable only within a set of private, cooperative sites, analogous to private IPv4 addresses (RFC 1918).
  - **Format:** Starts with `FC00::/7`.
  - **Purpose:** Used for communication within an organization without conflicting with other private networks and ensuring they are not mistakenly routed to the public Internet.
- **Link-Local Address:**
  - **What it is:** Automatically configured on every IPv6 interface, regardless of the presence of a router or DHCPv6 server.
  - **Format:** Starts with `FE80::/10`.
  - **Purpose:** Used only for communication with other nodes on the **same physical link** (e.g., for Neighbor Discovery Protocol, router advertisement). **They are not routable** beyond the link.

1.9.b Anycast

- **What it is:** An address assigned to **multiple interfaces** (typically routers/servers).
- **Purpose:** A packet sent to an Anycast address is routed to the **nearest** interface that has that address. This is often used for load-balancing and distributed services (like global DNS servers).

1.9.c Multicast

- **What it is:** An address assigned to a **group of interfaces**.
- **Format:** Starts with `FF00::/8`.
- **Purpose:** A packet sent to a Multicast address is delivered to **all interfaces** subscribed to that group (one-to-many communication).
- **Examples:** `FF02::1` (All nodes on the link), `FF02::2` (All routers on the link).

1.9.d Modified EUI-64

- **What it is:** A method for an IPv6 host to automatically generate the 64-bit Interface ID (host portion) of its IPv6 address.
- **How it works:**
  1. The host takes its 48-bit **MAC address**.
  2. It inserts the hexadecimal value `FFFE` in the middle (between the 3rd and 4th byte), resulting in a 64-bit value.
  3. It then **inverts the 7th bit** (the Universal/Local bit) of the 64-bit address.
- **Purpose:** Allows for stateless address auto-configuration (**SLAAC**) where the Interface ID is derived directly from a unique identifier (the MAC address).

1.10 Verify IP parameters for Client OS (Windows, Mac OS, Linux)

OS	Command to Verify IP Parameters	Key Information Displayed
Windows	`ipconfig` (or `ipconfig /all` for full details)	IPv4 Address, Subnet Mask, Default Gateway, DNS Servers.
Mac OS	`ifconfig` (older) or `ipconfig getifaddr en0` / **System Settings GUI**	IP Address, Subnet Mask, Gateway (Router), DNS.
Linux	`ip a` (or `ip addr show`) or `ifconfig` (older)	IP Address, Subnet Mask/Prefix, Link State, Broadcast Address.

- **Purpose:** To quickly confirm the host's IP configuration (address, mask, gateway) for troubleshooting connectivity issues.

1.11 Describe wireless principles

1.11.a Nonoverlapping Wi-Fi channels

- **What it is:** Specific frequency ranges within the 2.4 GHz and 5 GHz bands that do not interfere with each other when used simultaneously.
- **How it works:**
  - **2.4 GHz:** The channels 1, 6, and 11 are the only standard nonoverlapping 20 MHz channels in North America (each channel uses 5 adjacent channels, so 1, 6, and 11 are safely separated).
  - **5 GHz:** Offers many more nonoverlapping channels, reducing co-channel interference issues.
- **Why it matters:** Using nonoverlapping channels is **critical for wireless performance**. Overlapping channels cause **Co-Channel Interference (CCI)**, which forces devices to wait and retransmit, dramatically decreasing throughput.

1.11.b SSID (Service Set Identifier)

- **What it is:** The **human-readable name** of a Wi-Fi network (e.g., "MyOfficeWi-Fi").
- **Why it’s used:** Allows clients to easily identify and connect to a specific WLAN.

- **How it works:** Access Points broadcast the SSID in **beacon frames** to advertise the network's presence.

1.11.c RF (Radio Frequency)

- **What it is:** The electromagnetic waves used to transmit data wirelessly. Wi-Fi operates in the 2.4 GHz and 5 GHz unlicensed bands.
- **Why it matters:** RF signal strength, interference, and propagation characteristics determine the **range and throughput** of a wireless network.
- **Key Concepts:**
  - **Attenuation:** The loss of signal strength over distance or through materials (walls, etc.).
  - **Interference:** Signals from other sources (microwave ovens, other Wi-Fi networks) degrading the desired signal.

1.11.d Encryption

- **What it is:** The process of encoding wireless data to prevent unauthorized users from reading it.
- **Why it’s used:** Provides **confidentiality** for data transmitted over the air, which is easily intercepted.
- **How it works:** Uses a security protocol (WPA2, WPA3) and a **pre-shared key (PSK)** or **802.1X (EAP)** to generate encryption keys that scramble the data. Only devices with the correct key can decrypt the traffic.

1.12 Explain virtualization fundamentals (server virtualization, containers, and VRFs)

Server Virtualization

- **What it is:** The practice of running **multiple virtual machines (VMs)** on a single physical server (host).
- **How it works:** A **Hypervisor** (e.g., VMware ESXi, Microsoft Hyper-V) manages the host hardware and allocates resources (CPU, RAM, storage) to each VM. Each VM runs its own full **Operating System (OS)**.
- **Purpose:** Maximizes hardware utilization, reduces physical server count, lowers power consumption, and simplifies disaster recovery.

Containers

- **What it is:** A lightweight form of virtualization that packages an application and all its dependencies (libraries, configuration files) so it can run consistently across any environment.
- **How it works:** Containers **share the host operating system's kernel** but run in isolated user-space environments. They are much smaller and start faster than VMs. (Example: Docker).
- **Purpose:** Fast deployment, portability, and microservices architecture.

VRFs (Virtual Routing and Forwarding)

- **What it is:** A technology that allows a single physical router to host **multiple independent routing tables** simultaneously.
- **How it works:** Each VRF instance maintains a separate set of routing and forwarding tables, interfaces, and policies. It's like having multiple virtual routers in one box.
- **Purpose:** Provides **network segmentation** (similar to VLANs for switching) to keep traffic for different customers or departments completely separate, even if they share the same physical infrastructure (e.g., MPLS VPNs).

1.13 Describe switching concepts

1.13.a MAC learning and aging

- **MAC Learning:** The process by which a Layer 2 switch examines the **source MAC address** of an incoming frame and records it, along with the ingress port, in its **MAC Address Table (CAM table)**.
- **MAC Aging:** The process of removing MAC address entries from the MAC table after a certain period of inactivity (the aging time, typically 300 seconds). This ensures the table remains current and does not fill up with unused entries.

1.13.b Frame switching

- **What it is:** The fundamental process a Layer 2 switch uses to make a forwarding decision for an incoming Ethernet frame.
- **How it works:**
  1. The switch receives a frame.
  2. It records the **source MAC** and port (MAC learning).
  3. It checks the **destination MAC** address in the frame header against its MAC Address Table.
  4. If the destination MAC is found, the switch forwards the frame only out the **specific port** listed in the table (Unicast forwarding).

1.13.c Frame flooding

- **What it is:** The process where a switch forwards a frame out **all ports** in the same VLAN (except the port it was received on).
- **When it happens:**
  1. When the **destination MAC address is NOT in the MAC Address Table** (known as an **unknown unicast** frame).
  2. When the frame is a **broadcast** (destination MAC is `FFFF.FFFF.FFFF`).
  3. When the frame is a **multicast** (unless IGMP snooping is configured).
- **Consequence:** Too much flooding wastes bandwidth and can slow down the network.

1.13.d MAC address table (CAM Table)

- **What it is:** A memory table on the switch that stores the learned mapping of **MAC addresses** to **switch ports** and their respective **VLANs**. (Also known as the Content Addressable Memory (CAM)

Table).

- **Purpose:** It is the core mechanism that allows a switch to perform efficient, dedicated frame forwarding instead of flooding.
- **Verification (Cisco Switch):**

  Cisco CLI

```
Switch#show mac address-table
```

Domain 2.0 Network Access

2.1 Configure and verify VLANs (normal range) spanning multiple switches

VLAN (Virtual Local Area Network) is a logical segmentation of a Layer 2 switch network. It groups devices based on function, department, or application, regardless of their physical location.

- **What it is:** A broadcast domain that is isolated from other VLANs. A switch treats traffic within a VLAN as local and requires a Layer 3 device (router or L3 switch) to communicate between different VLANs.
- **Why it's used:**
  1. **Reduce Broadcast Domain Size:** Limits the scope of broadcast traffic, improving network efficiency.
  2. **Security:** Isolates sensitive data and departments.
  3. **Flexibility:** Groups users logically, simplifying moves, adds, and changes.
- **Why it matters:** VLANs are the foundation of modern, segmented enterprise networks.

Configuration (Normal Range VLANs 1-1005):

Cisco CLI

```
Switch>enable
Switch#configure terminal
Switch(config)#vlan 10
Switch(config-vlan)#name MARKETING
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name VOICE
Switch(config-vlan)#end
Switch#show vlan brief
```

2.1.a Access ports (data and voice)

- **What it is:** A switch port configured to belong to a **single VLAN** and intended for connecting an **end-user device** (like a PC, server, or IP phone).

- **Data Access Port:** Carries traffic for the connected endpoint, tagged with the port's assigned **Data VLAN**.
- **Voice Access Port:** Used for IP phones. It carries two types of traffic on one physical cable:
  1. **Data traffic** (from the PC connected to the phone) in the **Data VLAN**.
  2. **Voice traffic** (from the phone itself) in a separate **Voice VLAN** (often prioritized using QoS).

Configuration (Data and Voice):

Cisco CLI

```
Switch(config)#interface FastEthernet0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10      ! Data VLAN
Switch(config-if)#switchport voice vlan 20      ! Voice VLAN
Switch(config-if)#end
```

2.1.b Default VLAN

- **What it is: VLAN 1** is the pre-configured default VLAN on all Cisco switches.
- **Function/Purpose:**
  1. All ports are members of VLAN 1 by default.
  2. It is the default VLAN for control traffic like CDP, spanning tree, and management traffic unless explicitly changed.
- **Best Practice:** It is a **security best practice** to change all user/data ports and the management interface to a VLAN other than VLAN 1 to reduce potential attack surface.

2.1.c InterVLAN connectivity

- **What it is:** The process of allowing devices in **different VLANs (different subnets)** to communicate with each other.
- **Why it matters:** Since VLANs are separate broadcast domains, they are also separate IP subnets. **Routing** is required for communication between them.
- **How it works (Router-on-a-Stick):** A router connects to the switch via a single **trunk port**. The router interface is divided into logical sub-interfaces, one for each VLAN. The router performs the routing.
- **How it works (Layer 3 Switch):** The L3 switch has a **Switched Virtual Interface (SVI)** configured for each VLAN, giving it a default gateway address. The L3 switch handles the routing internally at high speed.

Configuration (SVI on a Layer 3 Switch):

Cisco CLI

```
Switch(config)#interface vlan 10
Switch(config-if)#ip address 192.168.10.1 255.255.255.0
```

```
Switch(config-if)#no shutdown
Switch(config-if)#interface vlan 20
Switch(config-if)#ip address 192.168.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config)#ip routing                ! Enable L3 routing globally
```

2.2 Configure and verify interswitch connectivity

2.2.a Trunk ports

- **What it is:** A switch port configured to carry traffic for **multiple VLANs** simultaneously.
- **Why it's used:** To connect switches to each other, or to connect a switch to a router (for InterVLAN routing).
- **How it works:** Frames traversing a trunk are **tagged** using a protocol (usually 802.1Q) to identify which VLAN they belong to.

Configuration (Trunk Port):

Cisco CLI

```
Switch(config)#interface GigabitEthernet0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20,99 ! Optional: specify allowed VLANs
Switch(config-if)#end
```

2.2.b 802.1Q

- **What it is:** The IEEE standard protocol for **VLAN tagging** on Ethernet trunk links. It is the industry standard.
- **How it works:** 802.1Q inserts a **4-byte tag field** into the Ethernet frame header. This tag includes the **12-bit VLAN ID (VID)**, which identifies the VLAN the frame belongs to.
- **Why it matters:** This tagging allows multiple VLAN frames to share the same physical link while maintaining segmentation.

2.2.c Native VLAN

- **What it is:** The VLAN assigned to a trunk port that will **not be tagged** when traversing the trunk link.
- **Function/Purpose:** Frames belonging to the Native VLAN are sent across the trunk **untagged**. When an untagged frame arrives on a trunk port, the switch assumes it belongs to the Native VLAN.
- **Best Practice:** The Native VLAN should be explicitly changed from VLAN 1 to an unused VLAN (e.g., VLAN 999) to prevent **VLAN hopping** attacks.
- **Configuration (Setting Native VLAN):**

  Cisco CLI

```
Switch(config-if)#switchport trunk native vlan 999
```

Note: Both sides of the trunk must match the Native VLAN configuration.

2.3 Configure and verify Layer 2 discovery protocols (Cisco Discovery Protocol and LLDP)

Protocol	CDP (Cisco Discovery Protocol)	LLDP (Link Layer Discovery Protocol)
Standard	**Cisco Proprietary**	**IEEE 802.1AB Standard**
Purpose	Gathers information about directly connected **Cisco** devices (model, capabilities, OS version, interface, etc.).	Gathers information about directly connected devices, **regardless of vendor**.
How it works	Sends periodic multicast frames to advertise its identity and status.	Sends periodic multicast frames to advertise its identity and status.
Why it matters	Essential for **network management and troubleshooting** (e.g., finding the switch port connected to a router, verifying duplex settings).	Necessary for **interoperability** in multi-vendor environments.

Configuration and Verification:

- **Global Configuration (CDP/LLDP):** Both are usually enabled by default. To enable/disable:

  Cisco CLI

  ```
  Switch(config)#cdp run
  Switch(config)#lldp run
  ```

- **Verification (CDP/LLDP):**

  Cisco CLI

  ```
  Switch#show cdp neighbors detail
  Switch#show lldp neighbors
  ```

2.4 Configure and verify (Layer 2/Layer 3) EtherChannel (LACP)

EtherChannel (or **Link Aggregation**) is a technology that groups multiple physical links into one logical link.

- **What it is:** A bundle of 2 to 8 parallel Ethernet links between two switches or a switch and a server.
- **Why it's used:**
  1. **Increased Bandwidth:** The throughput is the sum of the individual link speeds.
  2. **Redundancy:** If one link in the bundle fails, traffic automatically shifts to the remaining links (graceful failover).

3. **Spanning Tree (STP) Efficiency:** STP treats the bundle as a single logical link, avoiding the problem of blocking redundant paths.

- **Layer 2 EtherChannel:** Used for connecting switches (trunks or access links).
- **Layer 3 EtherChannel:** Used for connecting L3 switches or routers (routed ports) with a single IP address assigned to the logical interface.
- **LACP (Link Aggregation Control Protocol):** The **IEEE 802.3ad** standard protocol used to negotiate and manage the creation and state of an EtherChannel bundle.

Configuration (L2 EtherChannel using LACP):

Cisco CLI

```
Switch(config)#interface range GigabitEthernet0/1 - 2
Switch(config-if-range)#switchport mode trunk    ! Ensure link parameters match
Switch(config-if-range)#channel-protocol lacp
Switch(config-if-range)#channel-group 1 mode active ! 'active' enables LACP unconditionally
Switch(config-if-range)#exit

Switch(config)#interface Port-channel 1          ! Configure the logical interface
Switch(config-if)#switchport mode trunk
Switch(config-if)#end
```

Verification:

Cisco CLI

```
Switch#show etherchannel summary
Switch#show interfaces Port-channel 1
```

2.5 Interpret basic operations of Rapid PVST+ Spanning Tree Protocol (RSTP)

Spanning Tree Protocol (STP) / Rapid PVST+ is a Layer 2 protocol used to prevent **switching loops** in networks with redundant paths. Rapid PVST+ is the Cisco-proprietary version of **Rapid STP (802.1w)**, which converges much faster than the original 802.1D STP.

- **How it works:** Switches exchange **Bridge Protocol Data Units (BPDUs)** to determine the network topology and logically block redundant ports to create a **single, loop-free path**.

2.5.a Root port, root bridge (primary/secondary), and other port names

- **Root Bridge:** The single switch in a spanning tree domain that acts as the reference point for all path calculations. It is elected based on the lowest **Bridge ID (BID)** (priority + MAC address).

- **Root Port (RP):** The switch port on a non-Root Bridge that has the **lowest cost path** to the Root Bridge. Every non-Root switch has exactly one Root Port.
- **Designated Port (DP):** A port on a segment (link) that is closest to the Root Bridge in terms of cost. All ports on the Root Bridge are Designated Ports. Only Designated Ports can forward traffic.
- **Alternate/Backup Port:** A port that is blocked (in a Discarding state) but could immediately transition to a Forwarding state if the RP or DP fails.

2.5.b Port states and roles (RSTP)

RSTP significantly speeds up convergence by combining the 802.1D states and roles:

RSTP State	Old 802.1D State	Function
Discarding	Disabled, Blocking, Listening	The port discards incoming frames and does not learn MAC addresses. Prevents loops.
Learning	Learning	The port accepts incoming frames (for MAC learning) but does not forward frames yet.
Forwarding	Forwarding	The port sends and receives data frames (standard operation).

2.5.c PortFast

- **What it is:** A feature applied to access ports (connected to end devices like PCs/servers).
- **Purpose:** Allows the port to transition **immediately** from Blocking/Discarding to the **Forwarding** state, skipping the Listening and Learning states.
- **Why it matters:** Speeds up the time it takes for a PC to gain network access upon connection. **MUST NOT** be used on ports connecting to other switches, as it can cause temporary loops.

Configuration (PortFast):

Cisco CLI

```
Switch(config)#interface FastEthernet0/3
Switch(config-if)#spanning-tree portfast
```

2.5.d Root guard, loop guard, BPDU filter, and BPDU guard

These are Spanning Tree security and protection features:

Feature	Function/Purpose	How it Works
Root Guard	Enforces the Root Bridge location.	If a switch port configured with Root Guard receives a superior BPDU (suggesting a new, unauthorized Root Bridge), the port is put into a **Root-Inconsistent (Blocked)** state.

Feature	Function/Purpose	How it Works
Loop Guard	Protects against unidirectional link failures that can cause loops.	If a designated or root port stops receiving BPDUs, Loop Guard moves the port to a **Loop-Inconsistent (Blocked)** state instead of transitioning to Forwarding.
BPDU Filter	Stops the sending and receiving of BPDUs on an interface or globally.	When applied to a PortFast port, it prevents the switch from sending any BPDUs, useful for connecting to customer equipment.
BPDU Guard	Protects against unauthorized switches connecting to an access port.	If a port configured with BPDU Guard receives **any BPDU**, it immediately puts the port into an **err-disabled** (error-disabled) state. **Recommended for all PortFast ports.**

Configuration (BPDU Guard on interface):

Cisco CLI

```
Switch(config)#interface FastEthernet0/3
Switch(config-if)#spanning-tree bpduguard enable
```

2.6 Describe Cisco Wireless Architectures and AP modes

Cisco Wireless Architectures

Architecture	Description	Use Case
Centralized (Controller-Based)	The **WLC** handles management, control, and often the data traffic (centralized data plane). APs are "lightweight" and only provide RF connectivity.	**Large enterprises** where centralized policy enforcement, seamless roaming, and scalability are critical.
Distributed (Autonomous AP)	Each AP is an independent management and control entity. It handles its own configuration and forwards client traffic locally.	**Small offices/SOHO** environments with few APs and no need for centralized management or extensive roaming.
Cloud-Based (Cisco Meraki)	The WLC function is moved to the cloud. APs connect to the cloud management platform.	Deployments requiring minimal on-site infrastructure and simplified, web-based management.

AP Modes

AP Mode	Description
Local (Default)	AP connects to the WLC, tunnels all traffic (control and data) to the WLC via **CAPWAP**.

AP Mode	Description
Monitor	Dedicated to scanning for rogue devices, interferers, and RF statistics. Does not accept client connections.
FlexConnect	Allows APs at remote sites to handle local data switching (distributed data plane) when the WAN link to the WLC is down, but still receive central control.
Rogue Detector	APs listening for rogue APs in the network and reporting them to the WLC.
Mesh	APs that communicate wirelessly (backhaul) to extend coverage where wired connections are unavailable.

2.7 Describe physical infrastructure connections of WLAN components

- **AP to Switch:** The Access Point connects to the closest **Access Switch port**. This port is typically configured as a **trunk port** (to carry multiple VLANs: one for management, one or more for client traffic) and often enables **PoE** (Power over Ethernet).
- **WLC to Switch:** The Wireless LAN Controller connects to the **Distribution or Core Switch**. This connection is typically a high-speed **trunk port** (carrying all necessary client and management VLANs) or a **Link Aggregation Group (LAG)**/EtherChannel for redundancy and increased throughput.
- **Access Port:** The switch port where the AP is physically connected.
- **Trunk Ports:** The switch ports connecting the AP and WLC to the rest of the network, carrying multiple VLANs.
- **LAG (Link Aggregation Group):** An EtherChannel bundle used to connect the WLC to the network infrastructure for high bandwidth and redundancy.

2.8 Describe network device management access

Access Type	Description	Security	Best Practice
Console	Local physical access using a roll-over cable.	Highest initial security, relies on physical access control.	Used for initial setup and disaster recovery.
Telnet	Remote, terminal access over the network.	**Unencrypted.** Sends passwords and data in **clear text**.	**Avoid completely.** Only for legacy or non-critical environments.
SSH	Remote, terminal access over the network.	**Encrypted.** All communication (including authentication) is secure.	**Preferred method** for secure CLI management.

Access Type	Description	Security	Best Practice
HTTP	Web-based access.	**Unencrypted.** Sends passwords/data in clear text.	**Avoid.** Only for initial setup or local non-critical services.
HTTPS	Secure Web-based access using **TLS/SSL encryption**.	**Encrypted.** Highly secure web management.	**Preferred method** for secure GUI management.
TACACS+/RADIUS	Centralized Authentication, Authorization, and Accounting (AAA).	Uses a dedicated server (e.g., Cisco ISE) to verify user credentials and permissions for network device access.	**Mandatory for enterprise environments** for strong security, auditing, and compliance.
Cloud Managed	Device configuration and monitoring done via a centralized, often web-based, cloud dashboard (e.g., Meraki).	Security is handled by the cloud provider; access is via HTTPS/API.	Simplifies management across distributed networks.

2.9 Interpret the wireless LAN GUI configuration

A Wireless LAN (WLAN) GUI (typically on a WLC or cloud controller) allows for easy configuration of client connectivity settings.

- **WLAN Creation (SSID):** Define the **SSID name** that clients will see and connect to (e.g., "GuestNet").
- **Security Settings:** The most critical step. Select the appropriate encryption method:
  - **WPA2/WPA3 Personal (PSK):** Requires a pre-shared key (password) for all users. Simple for small deployments.
  - **WPA2/WPA3 Enterprise (802.1X):** Requires integration with a **RADIUS server** (e.g., TACACS+) to authenticate individual users with usernames and passwords. Required for enterprise security.
- **QoS Profiles:** Defines quality of service policies, often to **prioritize voice and video traffic** over best-effort data traffic.
- **Advanced Settings:** Includes configurations like:
  - **VLAN Assignment:** Assigns the WLAN to a specific VLAN.
  - **Client Exclusion/Isolation:** Prevents wireless clients from communicating directly with each other (Guest networks).
  - **Load Balancing/Band Steering:** Controls which AP and frequency band (2.4/5GHz) clients connect to for optimal performance.

Domain 3.0 IP Connectivity

3.1 Interpret the components of routing table

A **routing table** is a data structure stored in a router or Layer 3 switch that contains information about the routes to specific network destinations. This table is the core component that allows a router to make forwarding decisions.

Component	Definition	Function/Purpose
3.1.a Routing protocol code	A single character code indicating how the route was learned.	Identifies the **source** of the routing information (e.g., **C** for Connected, **S** for Static, **O** for OSPF, **D** for EIGRP).
3.1.b Prefix	The destination network address and its length (subnet mask/prefix length).	Defines the specific **destination network** being reached (e.g., `192.168.1.0/24`).
3.1.c Network mask	The 32-bit (IPv4) or 128-bit (IPv6) mask used to separate the network portion from the host portion.	Determines the **size** of the destination network. Often expressed as a CIDR notation prefix length (e.g., `/24`).
3.1.d Next hop	The IP address of the adjacent router to which the packet should be sent to continue its path toward the destination network.	Specifies the **exit point** from the current network segment. Can be an IP address or an exit interface.
3.1.e Administrative distance (AD)	A measure of the **trustworthiness** of the source of a routing protocol. A lower value is preferred.	Used when a router learns the **same route** from **multiple sources** (e.g., static and OSPF). The router installs the route with the lowest AD into the routing table.
3.1.f Metric	A value used by a routing protocol to determine the **best path** to a destination. The meaning varies by protocol (e.g., OSPF uses cost, RIP uses hop count).	Used when a router learns the **same route** from the **same routing protocol** source. The route with the lowest metric is preferred.
3.1.g Gateway of last resort	A specific address where the router sends packets when the destination network is **not found** in the routing table (i.e., a **default route**).	Allows a router to forward traffic to "the rest of the internet" or an upstream device without needing a route for every possible network.

Verification (Cisco Router):

Cisco CLI

```
Router#show ip route
```

3.2 Determine how a router makes a forwarding decision by default

A router follows a specific process to decide where to send an incoming packet:

3.2.a Longest prefix match

- **What it is:** The primary rule for forwarding decisions. The router compares the packet's destination IP address against all prefixes in its routing table.
- **How it works:** The router selects the route entry that has the **most specific match** (the longest prefix length/most bits matching).
- **Why it matters:** This ensures that traffic destined for a specific subnet is routed correctly, overriding a more general, less specific route.
- **Example:** If the destination is `192.168.1.10` and the table contains:
  1. `192.168.1.0/24` (24 bits match) -> Send via G0/1
  2. 192.168.1.0/26 (26 bits match) -> Send via G0/2

     The router chooses the /26 route because it is the longest (most specific) match.

3.2.b Administrative distance (AD)

- **What it is:** The **tie-breaker** when a router learns the **exact same destination prefix** from **different routing sources** (e.g., OSPF and EIGRP).
- **How it works:** The route source with the **lowest AD** is considered the most trusted and is installed in the routing table.
  - Connected: 0
  - Static: 1
  - EIGRP: 90
  - OSPF: 110
  - RIP: 120
- **Why it matters:** Guarantees that manually configured static routes are generally preferred over dynamically learned routes, and determines the preference between dynamic protocols.

3.2.c Routing protocol metric

- **What it is:** The final tie-breaker used when a router learns the **exact same destination prefix** from the **same routing source** (e.g., two OSPF routes).
- **How it works:** The route with the **lowest metric** (e.g., OSPF cost, EIGRP bandwidth/delay, RIP hop count) is installed in the routing table.
- **Why it matters:** Allows the routing protocol to select the most efficient path based on its internal calculation.

3.3 Configure and verify IPv4 and IPv6 static routing

Static Routing is when an administrator manually configures a router's path to a destination network.

3.3.a Default route

- **What it is:** A static route that directs all traffic for **unknown destinations** to a specified next-hop router. Also known as the **Gateway of Last Resort**.
- **Purpose:** The most common use is to direct traffic out to the Internet/WAN link.
- **Configuration (IPv4):**

  Cisco CLI

  ```
  Router(config)#ip route 0.0.0.0 0.0.0.0 [next-hop-ip | exit-interface]
  ```

- **Configuration (IPv6):**

  Cisco CLI

  ```
  Router(config)#ipv6 route ::/0 [next-hop-ipv6 | exit-interface]
  ```

3.3.b Network route

- **What it is:** A static route that specifies a path to a **specific network**.
- **Purpose:** Used for reaching specific, non-directly connected subnets.
- **Configuration (IPv4):**

  Cisco CLI

  ```
  Router(config)#ip route 172.16.1.0 255.255.255.0 10.1.1.2
  ```

  (Destination Network **172.16.1.0/24** is reached via next-hop router **10.1.1.2**)

3.3.c Host route

- **What it is:** A static route that specifies a path to a **single host IP address** (using a `/32` mask in IPv4 or a `/128` mask in IPv6).
- **Purpose:** Used when a router needs a specific path to a single device (e.g., a server) for security or special policy.
- **Configuration (IPv4):**

  Cisco CLI

  ```
  Router(config)#ip route 172.16.1.5 255.255.255.255 10.1.1.2
  ```

3.3.d Floating static

- **What it is:** A static route configured with a **higher Administrative Distance (AD)** than the primary route (usually learned dynamically).
- **Purpose:** To provide a **backup path** for a destination network. It remains dormant unless the primary route fails, at which point the floating static route's higher AD makes it the best available path.
- **Configuration (IPv4 - AD of 5 is higher than OSPF's default of 110, but in this case, AD must be greater than primary):**
  - *Assume primary route is OSPF (AD 110).*

Cisco CLI

```
Router(config)#ip route 172.16.1.0 255.255.255.0 10.1.1.2 150
```

(AD is set to **150** - only active if OSPF fails or is removed).

Verification (All Static Routes):

Cisco CLI

```
Router#show ip route static
```

3.4 Configure and verify single area OSPFv2

OSPFv2 (Open Shortest Path First, Version 2) is a link-state dynamic routing protocol used for IPv4 networks. It is highly scalable and efficient.

- **Link-State:** Routers share a complete map of the entire network topology, allowing them to independently calculate the best path.
- **Area:** OSPF uses a two-tier hierarchy with **Area 0 (Backbone)** and other regular areas. CCNA focuses on **Single Area OSPF** (everything is in Area 0).

3.4.a Neighbor adjacencies

- **What it is:** The establishment of a relationship between two directly connected OSPF routers after exchanging **Hello packets** and **Link State Advertisements (LSAs)**.
- **How it works:** Routers must agree on parameters (e.g., Hello/Dead intervals, Subnet mask, Area ID) to become **Full** neighbors.
- **Verification:**
  Cisco CLI

  ```
  Router#show ip ospf neighbor
  ```

3.4.b Point-to-point (P2P)

- **What it is:** OSPF operating mode over a dedicated link between two routers (e.g., a serial link or a high-speed fiber link).
- **Characteristic:** No need for a DR/BDR election, as there are only two routers. Fast convergence.

3.4.c Broadcast (DR/BDR selection)

- **What it is:** OSPF operating mode on multi-access networks (like Ethernet LANs) where multiple routers are connected to a single segment.
- **DR/BDR Selection:** A **Designated Router (DR)** and a **Backup Designated Router (BDR)** are elected on each multi-access segment.
  - **DR Function:** The designated router is responsible for exchanging routing updates with all other routers on that segment, minimizing the number of adjacencies and reducing LSA flooding.

    - **BDR Function:** Stands by to take over the role of the DR immediately if the DR fails.
- **Election Criteria:** Highest **OSPF Priority** (default 1), then highest **Router ID**.

3.4.d Router ID (RID)

- **What it is:** A unique 32-bit identifier (formatted like an IPv4 address) used by an OSPF router to represent itself in the OSPF domain.
- **How it's chosen (in order of preference):**
    1. Manually configured using the `router-id` command.
    2. Highest IP address of any active **Loopback interface**.
    3. Highest IP address of any active **physical interface**.
- **Why it matters:** Critical for forming adjacencies and identifying the source of LSAs.

Configuration (Single Area OSPFv2):

Cisco CLI

```
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1          ! Set the RID manually
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0  ! Advertises
network 192.168.1.0/24
Router(config-router)#network 10.1.1.0 0.0.0.3 area 0      ! Advertises P2P
link 10.1.1.0/30
Router(config-router)#end
```

3.5 Describe the purpose, functions, and concepts of first hop redundancy protocols (FHRP)

- **What it is:** A group of protocols that prevent network outages when a default gateway (first-hop router) fails.
- **Purpose:** To provide a **highly available** default gateway for host devices (PCs, servers) on a LAN.
- **How it works:** FHRPs use a **virtual IP (VIP)** and a **virtual MAC address** shared between two or more physical routers. The hosts use the VIP as their default gateway.
    - One router is **Active** (owner of the VIP/VMAC) and forwards traffic.
    - The other router is **Standby/Listen** and constantly monitors the Active router.
    - If the Active router fails, the Standby router instantly takes over the VIP/VMAC, creating a **transparent failover** for the hosts.

FHRP Examples:

1. **HSRP (Hot Standby Router Protocol):** Cisco proprietary. Uses Active/Standby roles.
2. **VRRP (Virtual Router Redundancy Protocol):** Open standard (RFC 5798). Uses Master/Backup roles.

3. **GLBP (Gateway Load Balancing Protocol):** Cisco proprietary. Allows **load sharing** across multiple physical routers, simultaneously using multiple virtual MAC addresses.

🛡️ First Hop Redundancy Protocols (FHRP) Configuration

FHRPs ensure that if the default gateway (first-hop router) fails, a standby router immediately takes over its IP and MAC addresses, providing transparent failover for hosts.

1. HSRP (Hot Standby Router Protocol)

HSRP is a Cisco proprietary protocol that uses **Active** and **Standby** roles, sharing a **Virtual IP Address (VIP)** and a **Virtual MAC Address**.

Goal

Provide default gateway redundancy for the subnet **192.168.10.0/24**.

- **Router 1 (R1):** Primary/Active router.
- **Router 2 (R2):** Secondary/Standby router.
- **Virtual IP (VIP):** `192.168.10.254` (The host's default gateway).

Step-by-Step Configuration

➡️ Router 1 (R1) - Active

1. **Enter the interface configuration mode:**

   Cisco CLI

   ```
   R1#configure terminal
   R1(config)#interface GigabitEthernet0/0
   ```

2. **Configure the physical IP address:**

   Cisco CLI

   ```
   R1(config-if)#ip address 192.168.10.1 255.255.255.0
   ```

3. **Enable HSRP and assign the Virtual IP (VIP):**

   - The `standby 10` is the **HSRP Group Number**. This number must match on all routers in the group.

   Cisco CLI

   ```
   R1(config-if)#standby 10 ip 192.168.10.254
   ```

4. **Set the Priority (Highest is Active):**

   - The default priority is 100. Setting R1 to 150 ensures it becomes Active.

   Cisco CLI

```
R1(config-if)#standby 10 priority 150
```

5. **Enable Preemption:**
   - This forces R1 to immediately take over the Active role if it reboots or its priority increases, even if R2 is currently Active.

   Cisco CLI

```
R1(config-if)#standby 10 preempt
R1(config-if)#end
```

➡️ **Router 2 (R2) - Standby**

1. **Enter the interface configuration mode:**

   Cisco CLI

```
R2#configure terminal
R2(config)#interface GigabitEthernet0/0
```

2. **Configure the physical IP address:**

   Cisco CLI

```
R2(config-if)#ip address 192.168.10.2 255.255.255.0
```

3. **Enable HSRP and assign the Virtual IP (VIP):**
   - The HSRP group number and VIP must match R1's configuration.

   Cisco CLI

```
R2(config-if)#standby 10 ip 192.168.10.254
```

4. **Set the Priority:**
   - Default priority of 100 is lower than R1's 150, so R2 will be Standby.

   Cisco CLI

```
R2(config-if)#standby 10 priority 100
R2(config-if)#end
```

Verification

Cisco CLI

```
R1#show standby brief
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Gi0/0       10   150 P Active  local           192.168.10.2
192.168.10.254

R2#show standby brief
```

```
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Gi0/0       10   100   Standby 192.168.10.1    local
192.168.10.254
```

---

2. VRRP (Virtual Router Redundancy Protocol)

VRRP is an industry standard (IEEE) protocol similar to HSRP. It uses **Master** and **Backup** roles, sharing a **Virtual IP (VIP)** and a **Virtual MAC Address**.

Goal

Provide default gateway redundancy for the subnet **192.168.20.0/24**.

- **Router 3 (R3):** Primary/Master router.
- **Router 4 (R4):** Secondary/Backup router.
- **Virtual IP (VIP):** `192.168.20.254`.

Step-by-Step Configuration

➡️ **Router 3 (R3) - Master**

1. **Enter the interface configuration mode:**

   Cisco CLI

   ```
   R3#configure terminal
   R3(config)#interface GigabitEthernet0/1
   ```

2. **Configure the physical IP address:**

   Cisco CLI

   ```
   R3(config-if)#ip address 192.168.20.1 255.255.255.0
   ```

3. **Enable VRRP and assign the Virtual IP (VIP):**

   - The `vrrp 20` is the **VRRP Group Number**.

   Cisco CLI

   ```
   R3(config-if)#vrrp 20 ip 192.168.20.254
   ```

4. **Set the Priority (Highest is Master):**

   - Setting R3 to 150 ensures it becomes the Master.

   Cisco CLI

   ```
   R3(config-if)#vrrp 20 priority 150
   ```

5. **Enable Preemption:**

   Cisco CLI

```
R3(config-if)#vrrp 20 preempt
R3(config-if)#end
```

➡️ **Router 4 (R4) - Backup**

1. **Enter the interface configuration mode:**

   Cisco CLI

   ```
   R4#configure terminal
   R4(config)#interface GigabitEthernet0/1
   ```

2. **Configure the physical IP address:**

   Cisco CLI

   ```
   R4(config-if)#ip address 192.168.20.2 255.255.255.0
   ```

3. **Enable VRRP and assign the Virtual IP (VIP):**

   Cisco CLI

   ```
   R4(config-if)#vrrp 20 ip 192.168.20.254
   ```

4. **Set the Priority:**

   - Default priority of 100 is sufficient to remain Backup.

   Cisco CLI

   ```
   R4(config-if)#vrrp 20 priority 100
   R4(config-if)#end
   ```

Verification

Cisco CLI

```
R3#show vrrp brief
Interface   Grp  State   Master IP       Virtual IP      VRRP Group
Gi0/1       20   Master  Local           192.168.20.254  20

R4#show vrrp brief
Interface   Grp  State   Master IP       Virtual IP      VRRP Group
Gi0/1       20   Backup  192.168.20.1    192.168.20.254  20
```

---

3. GLBP (Gateway Load Balancing Protocol)

GLBP is a Cisco proprietary protocol that not only provides redundancy but also allows for **load sharing** (traffic balancing) across up to four physical routers simultaneously.

- **AVG (Active Virtual Gateway):** The router that handles the virtual IP address and assigns the virtual MAC addresses.

- **AVF (Active Virtual Forwarder):** The router that actually forwards the traffic for a specific virtual MAC address.

Goal

Provide default gateway redundancy and load sharing for the subnet **192.168.30.0/24**.

- **Router 5 (R5):** Primary/AVG and AVF for MAC 1.
- **Router 6 (R6):** Secondary/AVF for MAC 2.
- **Virtual IP (VIP):** `192.168.30.254`.

Step-by-Step Configuration

➡️ Router 5 (R5) - AVG (Virtual IP Owner)

1. **Enter the interface configuration mode:**

   Cisco CLI

   ```
   R5#configure terminal
   R5(config)#interface GigabitEthernet0/2
   ```

2. **Configure the physical IP address:**

   Cisco CLI

   ```
   R5(config-if)#ip address 192.168.30.1 255.255.255.0
   ```

3. **Enable GLBP and assign the Virtual IP (VIP):**

   - The `glbp 30` is the **GLBP Group Number**.

   Cisco CLI

   ```
   R5(config-if)#glbp 30 ip 192.168.30.254
   ```

4. **Set the Priority (Highest is AVG):**

   - Priority of 150 ensures R5 is the Active Virtual Gateway (AVG).

   Cisco CLI

   ```
   R5(config-if)#glbp 30 priority 150
   ```

5. **Enable Preemption:**

   - Ensures R5 takes back the AVG role if it recovers.

   Cisco CLI

   ```
   R5(config-if)#glbp 30 preempt
   R5(config-if)#end
   ```

➡️ Router 6 (R6) - Backup AVG / AVF

1. **Enter the interface configuration mode:**

   Cisco CLI

   ```
   R6#configure terminal
   R6(config)#interface GigabitEthernet0/2
   ```

2. **Configure the physical IP address:**

   Cisco CLI

   ```
   R6(config-if)#ip address 192.168.30.2 255.255.255.0
   ```

3. **Enable GLBP and assign the Virtual IP (VIP):**

   Cisco CLI

   ```
   R6(config-if)#glbp 30 ip 192.168.30.254
   ```

4. **Set the Priority:**

   - Default priority of 100 is sufficient.

   Cisco CLI

   ```
   R6(config-if)#glbp 30 priority 100
   R6(config-if)#end
   ```

Verification

Cisco CLI

```
R5#show glbp brief
Interface   Grp  State    Pri Active Virtual Router  Standby Virtual Router
Gi0/2       30   Active   150 192.168.30.1           192.168.30.2
             -- Group 30
             State is Active
             Virtual IP address is 192.168.30.254
             Forwarder 1: State is Active (primary)
             Forwarder 2: State is Active
```

Interpretation: R5 is the **AVG** (Active) and also one of the **Forwarders** (AVF). R6 is the **Standby** (for the AVG role) and also a Forwarder. The AVG (R5) is intelligently assigning different Virtual MAC addresses to clients via ARP replies, causing traffic to be split between R5 and R6.

Domain 4.0 IP Services

4.1 Configure and verify inside source NAT using static and pools

Network Address Translation (NAT) is a process that translates one IP address (usually a private, internal address) into another IP address (usually a public, external address) as the traffic traverses a

boundary device (typically a router or firewall).

- **Purpose:** To conserve public IPv4 addresses and provide a layer of security by hiding internal network addresses from the outside world.
- **Terms:**
  - **Inside Local (IL):** The private IP address of the source host.
  - **Inside Global (IG):** The public IP address used by the source host when communicating with the outside.
  - **Outside Local (OL):** The IP address of the destination host as it appears to the inside network.
  - **Outside Global (OG):** The real public IP address of the destination host.

Static NAT (One-to-One)

- **What it is:** A permanent mapping of a single **Inside Local** address to a single **Inside Global** address.
- **Use Case:** Allowing external users to access an internal resource (e.g., a Web Server or Email Server) using a publicly known IP address.

Configuration (Static NAT)

- **Goal:** Map internal server `192.168.1.10` to public address `203.0.113.10`.

1. **Define the NAT mapping:**

   Cisco CLI

   ```
   Router(config)#ip nat inside source static 192.168.1.10 203.0.113.10
   ```

2. **Identify the NAT interfaces:**

   Cisco CLI

   ```
   Router(config)#interface GigabitEthernet0/0
   Router(config-if)#ip nat inside      ! Interface facing the internal
   (private) network
   Router(config-if)#exit
   Router(config)#interface GigabitEthernet0/1
   Router(config-if)#ip nat outside     ! Interface facing the external
   (public) network
   Router(config-if)#end
   ```

Dynamic NAT with Pools (One-to-Many or Many-to-Many)

- **What it is:** Allows multiple **Inside Local** addresses to share a limited pool of **Inside Global** addresses. When a host initiates traffic, NAT selects an available public address from the pool.
- **Use Case:** When a company has more private hosts than available public IP addresses (though **PAT** is more common for this, see note below).

Configuration (Dynamic NAT with Pools)

- **Goal:** Allow hosts in the `192.168.1.0/24` subnet to use public addresses from the range `203.0.113.100` to `203.0.113.110`.

1. **Define the range of private addresses to be translated (Access List):**

   Cisco CLI

   ```
   Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
   ```

2. **Define the pool of public (Inside Global) addresses:**

   Cisco CLI

   ```
   Router(config)#ip nat pool PUBLIC_POOL 203.0.113.100 203.0.113.110
   netmask 255.255.255.0
   ```

3. **Tie the Access List to the Pool:**

   Cisco CLI

   ```
   Router(config)#ip nat inside source list 1 pool PUBLIC_POOL
   ```

4. **Identify the NAT interfaces** (same as Static NAT):

   Cisco CLI

   ```
   Router(config)#interface GigabitEthernet0/0
   Router(config-if)#ip nat inside
   Router(config-if)#exit
   Router(config)#interface GigabitEthernet0/1
   Router(config-if)#ip nat outside
   Router(config-if)#end
   ```

> Note on PAT (Port Address Translation): The most common form of Dynamic NAT is overloading, or PAT, where multiple private hosts share a single public IP address using different port numbers. The configuration for PAT is done by adding the keyword overload to the end of the NAT command:
>
> ip nat inside source list 1 interface GigabitEthernet0/1 overload

Verification (NAT)

Cisco CLI

```
Router#show ip nat translations  ! View the active translation table
Router#clear ip nat translation * ! Clear all current NAT entries
```

4.2 Configure and verify NTP operating in a client and server mode

Network Time Protocol (NTP) is used to synchronize the clocks of computer systems over a network. Accurate time is crucial for logging, troubleshooting, security, and ensuring proper sequencing of events.

- **Stratum:** A measure of the distance from the reference clock (Stratum 0). A Stratum 1 server connects directly to a reference clock; a Stratum 2 server gets time from a Stratum 1 server, and so on.

Step-by-Step Configuration

➡️ Router/Switch as an NTP Client

The device synchronizes its clock with an upstream NTP Server.

1. **Enter configuration mode and set the server:**

   Cisco CLI

   ```
   Device(config)#ntp server 192.168.100.1 source GigabitEthernet0/0
   ```

   (The `source` command ensures the router uses a specific local interface for the NTP packets, which is good practice).

2. **Enable the service (usually on by default):**

   Cisco CLI

   ```
   Device(config)#ntp enable
   Device(config)#end
   ```

➡️ Router/Switch as an NTP Server

The device receives time from an upstream server and then provides that synchronized time to other devices (clients) on its network.

1. **Configure the device as an NTP client (see steps above) to get synchronized time.**
2. **Configure the device to be an NTP Master (Server):**
   - The command specifies the **Stratum level** (2-15) this device will advertise.

   Cisco CLI

   ```
   Device(config)#ntp master 4  ! This device will advertise itself as a
   Stratum 4 server
   Device(config)#end
   ```

Verification (NTP)

Cisco CLI

```
Device#show ntp status     ! Shows synchronization status (e.g.,
synchronized, stratum)
Device#show ntp associations ! Shows which server/peer it is connected to
Device#show clock detail   ! Displays the current time and synchronization
source
```

4.3 Explain the role of DHCP and DNS within the network

DHCP (Dynamic Host Configuration Protocol)

- **Role:** An application layer protocol that **automatically assigns IP configuration parameters** to client devices.
- **Function:** Eliminates the need for manual, error-prone static configuration. It manages a pool of addresses and leases them out.
- **Purpose (DORA Process):**
  1. **Discover:** Client broadcasts a request for an IP address.
  2. **Offer:** DHCP server sends an offer with an IP address and parameters.
  3. **Request:** Client accepts the offer.
  4. **Acknowledge:** DHCP server sends the final ACK, confirming the lease.
- **Parameters Assigned:** IP address, subnet mask, default gateway, and DNS server addresses.

DNS (Domain Name System)

- **Role:** The fundamental protocol that translates **human-readable domain names** (e.g., `www.google.com`) into **machine-readable IP addresses** (e.g., `142.250.69.196`).
- **Function:** Essential for navigation on the Internet and internal networks.
- **Purpose:** The DNS hierarchy provides a decentralized, scalable, and resilient way to manage the mapping of names to addresses.

4.4 Explain the function of SNMP in network operations

Simple Network Management Protocol (SNMP) is an application layer protocol used to collect, organize, and modify information about network devices to monitor their status and behavior.

- **Function:** Allows a central **Network Management Station (NMS)** to query and manage network devices (routers, switches, servers) remotely.
- **Purpose:** Provides visibility into the health and performance of the network for troubleshooting, capacity planning, and proactive maintenance.
- **Key Components:**
  1. **SNMP Manager (NMS):** Software (e.g., Nagios, SolarWinds) that runs on a central server to collect and process data.
  2. **SNMP Agent:** Software running on the network device (router, switch) that collects local data and responds to manager requests.
  3. **MIB (Management Information Base):** A hierarchical database of variables that the agent manages. The manager queries or modifies these variables.
  4. **Community String:** A plaintext "password" used for authentication between the manager and the agent.
     - **`read-only (ro)`:** Allows the manager to retrieve information.

- **`read-write (rw)`:** Allows the manager to both retrieve and modify configuration.

5. **Trap/Inform:** Messages sent **unsolicited** by the Agent to the Manager to alert about a critical event (e.g., link down).

Configuration (Basic SNMP)

- **Goal:** Configure a Cisco device to allow SNMP queries from NMS `10.1.1.5` using the read-only community string `public_ro`.

Cisco CLI

```
Device(config)#snmp-server community public_ro ro 10.1.1.5
Device(config)#snmp-server contact Network Admin Team
Device(config)#end
```

4.5 Describe the use of syslog features, including facilities and severity levels

Syslog is a standard protocol used to send system log or event messages from a device to a centralized server (Syslog Server).

- **Use:** Centralizing logs makes it easier to review events, troubleshoot problems, detect security threats, and perform network auditing.

Facilities

- **What it is:** Identifies the **source process or module** that generated the message.
- **Common Examples:**
  - `local7`: Often used for router/switch commands.
  - `kernel`: Operating system core messages.
  - `security/auth`: Security and authentication events.

Severity Levels

- **What it is:** A numerical code (0 to 7) indicating the **criticality** of the event. Lower numbers are more severe.

Level (Number)	Keyword	Description
0	**emergencies**	System is unusable. (Highest priority)
1	**alerts**	Immediate action required.
2	**critical**	Critical conditions (e.g., hardware failure).
3	**errors**	Error conditions.
4	**warnings**	Warning conditions (e.g., link flapping).

Level (Number)	Keyword	Description
5	**notifications**	Normal, but significant conditions (e.g., interface up/down).
6	**informational**	Routine information (e.g., successful login).
7	**debugging**	Detailed information for debugging software. (Lowest priority)

Configuration (Basic Syslog)

- **Goal:** Send logs to server `10.1.1.6` and set the console logging to show only critical events and above.

Cisco CLI

```
Router(config)#logging 10.1.1.6
Router(config)#logging console critical ! Sets the console display threshold
Router(config)#logging trap warnings   ! Sends warnings and higher to the
server
Router(config)#end
```

4.6 Configure and verify DHCP client and relay

DHCP Client Configuration

- **What it is:** The configuration of an interface to dynamically obtain its IP configuration from a DHCP server.
- **Use Case:** Typical configuration for endpoints (PCs, Laptops) or WAN interfaces on a router connected to an ISP.

Configuration (Router Interface as DHCP Client)

Cisco CLI

```
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address dhcp   ! Instructs the router to use DHCP
Router(config-if)#no shutdown
Router(config-if)#end
```

DHCP Relay Agent

- **What it is:** A router or Layer 3 switch configured to forward DHCP broadcast requests from a client network to a DHCP server located on a **different network segment**.
- **Why it's needed:** DHCP uses broadcasts, which do not cross router boundaries. The relay agent converts the broadcast into a unicast message.

Configuration (Router as DHCP Relay Agent)

- **Goal:** Relay DHCP requests received on interface G0/1 to the DHCP server at `192.168.200.5`.

1. **Enter the client-facing interface:**

   Cisco CLI

   ```
   Router(config)#interface GigabitEthernet0/1
   ```

2. **Configure the DHCP relay command (The `ip helper-address`):**

   Cisco CLI

   ```
   Router(config-if)#ip helper-address 192.168.200.5
   Router(config-if)#end
   ```

Verification (DHCP)

Cisco CLI

```
Router#show ip interface brief  ! Check if the interface got an IP address
Router#show ip dhcp binding     ! View the IP addresses leased by the server
(if applicable)
```

4.7 Explain the forwarding per-hop behavior (PHB) for QoS

Quality of Service (QoS) refers to the set of mechanisms used to manage network traffic to ensure critical applications (like Voice and Video) receive the necessary bandwidth, low latency, and low jitter.

Per-Hop Behavior (PHB): The action taken by a network device (router or switch) on a packet as it traverses the network, based on the packet's markings.

QoS Mechanism	Description	Location	Purpose
Classification	Identifying and grouping traffic based on fields like source/destination IP, port number, or protocol.	Edge (Ingress)	First step in QoS; determines what priority a packet gets.
Marking	Placing an identifier (a tag) on the packet header to convey its priority to downstream devices.	Edge (Ingress)	Common markings are **CoS** (Layer 2) and **DSCP** (Layer 3 - recommended).
Queuing	Buffering packets and scheduling them for transmission, typically based on their markings.	Congested Devices	Manages the flow when traffic exceeds link capacity, prioritizing critical traffic.
Congestion	The state where a network resource (buffer) is overwhelmed and packets begin to drop.	Congested Devices	QoS mechanisms like **Tail Drop** or **WRED** are used to handle this.

QoS Mechanism	Description	Location	Purpose
Policing	Dropping or remarking (lowering the priority) of traffic that **exceeds** a defined rate.	Edge (Ingress)	Used to enforce contracts (SLA); acts like a strict traffic cop.
Shaping	Buffering excess traffic and sending it out in smoothed bursts to conform to a target rate.	Edge (Egress)	Used to smooth traffic flow; less aggressive than policing.

4.8 Configure network devices for remote access using SSH

Secure Shell (SSH) is a cryptographic protocol used for secure remote login and command-line execution over an unsecure network. It is the mandatory replacement for the unencrypted Telnet protocol.

Step-by-Step Configuration (SSH)

- **Goal:** Enable SSH access for local users.

1. **Configure a Hostname and Domain Name:** (Required to generate cryptographic keys)

   Cisco CLI

   ```
   Router(config)#hostname R1
   R1(config)#ip domain name example.com
   ```

2. **Generate the RSA Crypto Key Pair:** (Used for encryption)
   - A key size of 1024 or 2048 is recommended.

   Cisco CLI

   ```
   R1(config)#crypto key generate rsa modulus 1024
   ```

3. **Create a Local User Account:** (SSH requires local or AAA authentication)

   Cisco CLI

   ```
   R1(config)#username admin secret CiscoCCNA
   ```

4. **Configure the VTY lines (Virtual Teletype):**
   - VTY lines are used for inbound, remote terminal connections.

   Cisco CLI

   ```
   R1(config)#line vty 0 4             ! Affects VTY lines 0 through 4 (5
   total sessions)
   R1(config-line)#transport input ssh  ! Allows only SSH connections
   (disables Telnet)
   R1(config-line)#login local         ! Uses the local username/secret
   ```

```
database for authentication
R1(config-line)#end
```

Verification (SSH)

Cisco CLI

```
R1#show ip ssh
R1#show users
```

4.9 Describe the capabilities and functions of TFTP/FTP in the network

Protocol	TFTP (Trivial File Transfer Protocol)	FTP (File Transfer Protocol)
Port(s)	UDP Port 69	TCP Ports 20 (data) and 21 (control)
Transport	**UDP** (Unreliable, connectionless)	**TCP** (Reliable, connection-oriented)
Authentication	**None** (Anonymous access only)	**Requires** username and password
Security	None (sends data in plaintext)	None (sends data and credentials in plaintext)
Primary Use	Transferring **small** configuration files, Cisco IOS images, or boot files to devices.	General-purpose transfer of large files between systems, commonly used for public downloads.
Function	Extremely simple, low overhead. Cannot list directories or perform advanced file operations.	Full-featured file transfer: supports directory listing, renaming, deletion, and robust error checking.

Configuration (TFTP/FTP)

- Network devices use these protocols via the CLI for file management:

  Cisco CLI

  ```
  Router#copy running-config tftp:
  Router#copy tftp: flash:IOS_FILE.bin
  ```

Domain 5.0 Security Fundamentals

5.1 Define key security concepts (threats, vulnerabilities, exploits, and mitigation techniques)

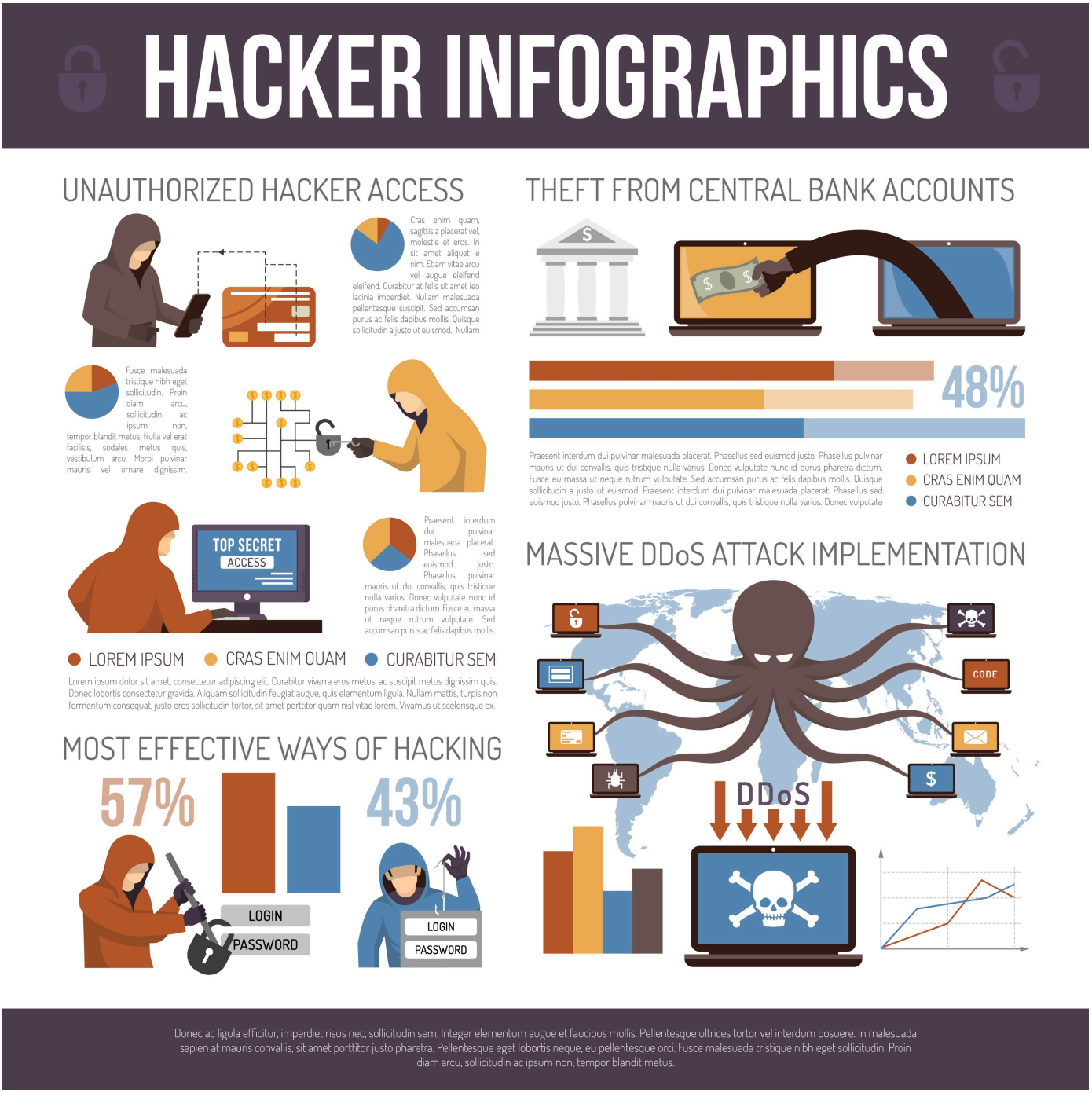
Concept	Definition	Purpose/Why it Matters
Threat	A potential danger that could compromise an asset (e.g., data, network device).	Represents the potential for harm (e.g., a malware attack, a disgruntled employee).
Vulnerability	A **weakness** in a system, design, or implementation that can be exploited by a threat.	The specific hole or flaw (e.g., unpatched software, weak password, misconfigured firewall).
Exploit	A piece of software, data, or command sequence that takes **advantage of a vulnerability** to cause unintended or unanticipated behavior.	The mechanism used by an attacker to successfully compromise a system.
Mitigation Technique	The method or control used to **reduce the probability or impact** of a threat or vulnerability.	The solution implemented to secure the environment (e.g., patching, strong passwords, ACLs).

Export to Sheets

5.2 Describe security program elements (user awareness, training, and physical access control)

Element	Description	Purpose
User Awareness	Educating end-users about security risks and best practices (e.g., phishing emails, safe browsing, reporting incidents).	Users are the **weakest link** in the security chain. Awareness turns users into the first line of defense.
Training	In-depth instruction for specific roles (e.g., teaching network administrators how to configure secure protocols like SSH and access lists).	Ensures personnel have the **technical skills** required to implement and maintain security controls.
Physical Access Control	Security measures to restrict access to network devices and data centers (e.g., locked server rooms, video surveillance, badge readers).	Prevents unauthorized users from gaining **direct access** to infrastructure, which is a common point of compromise.

Export to Sheets



Shutterstock

5.3 Configure and verify device access control using local passwords

The most basic security measure is protecting management access to the console and VTY lines.

Step-by-Step Configuration (Local Password Security)

1. **Configure Console Port Access:**

   Cisco CLI

```
Router(config)#line console 0
Router(config-line)#password cisco
```

```
Router(config-line)#login
Router(config-line)#end
```

This requires the password "cisco" to access the console, but the password is sent in plaintext.

2. **Configure VTY (Telnet/SSH) Access:**

   Cisco CLI

   ```
   Router(config)#line vty 0 4
   Router(config-line)#password cisco
   Router(config-line)#login
   Router(config-line)#end
   ```

3. **Encrypt the Passwords:** (Prevents passwords from being seen in the configuration file)

   Cisco CLI

   ```
   Router(config)#service password-encryption
   ```

   This command encrypts all passwords configured using the `password` command (Type 7 encryption, which is easily reversible).

4. **Configure a Secure/Secret Password:** (Uses a stronger Type 5 or Type 8 encryption)

   Cisco CLI

   ```
   Router(config)#enable secret VeryStrongPass!
   ```

   *This is used to protect the **Privileged EXEC Mode** (`enable` access).*

5.4 Describe security password policy elements

A comprehensive password policy is essential for mitigating brute-force and dictionary attacks.

Element	Description	Why it Matters
Management	Defining roles and processes for password changes, resets, and storage.	Ensures passwords are not shared, easily discoverable, or reusable.
Complexity	Requirements for minimum length, mixing of character types (upper, lower, number, symbol), and avoiding dictionary words.	Increases the entropy (randomness) of the password, making it harder to guess or crack.
Password Alternatives	Methods replacing or supplementing the traditional static password.	Provides stronger security against phishing and credential theft.
Multifactor Authentication (MFA)	Requires two or more verification factors (e.g., password **and** a one-time code from a phone).	Even if the password is stolen, the attacker cannot log in without the second factor.
Certificates	Digital documents that verify the identity of an entity (user or device)	Used in SSH, VPNs, and HTTPS for secure

Element	Description	Why it Matters
	using Public Key Infrastructure (PKI).	authentication and non-repudiation.
Biometrics	Authentication using unique biological traits (e.g., fingerprint, retina scan).	Difficult to forge or steal, providing a high degree of assurance for physical and logical access.

Export to Sheets

5.5 Describe IPsec remote access and site-to-site VPNs

VPN (Virtual Private Network) provides a secure, encrypted tunnel over an unsecure network (like the Internet). **IPsec (Internet Protocol Security)** is the standard protocol suite used to secure IP communication.

VPN Type	Description	Use Case
Site-to-Site VPN	Connects **two separate fixed networks** (e.g., a corporate headquarters and a branch office) over the Internet, making them appear as one private network.	Interconnecting offices permanently. Devices behind the VPN are unaware they are using a VPN.
Remote Access VPN	Connects a **single remote user** (e.g., an employee working from home) back to the corporate network.	Enabling secure access to internal resources for traveling or remote workers. Requires VPN client software.

Export to Sheets

IPsec Components

- **IKE (Internet Key Exchange):** Used for **authentication** (verifying identity) and **key management** (creating and refreshing shared encryption keys).
- **IPsec Tunnel Mode:** Encrypts the entire original IP packet (header and data). Used for most VPNs.
- **Security Protocols (AH/ESP):**
  - **AH (Authentication Header):** Provides **authentication** and **integrity** (ensures data hasn't been tampered with). Does **not** provide encryption.
  - **ESP (Encapsulating Security Payload):** Provides **confidentiality (encryption)** and often integrity/authentication. **Most common** choice for IPsec VPNs.

5.6 Configure and verify access control lists (ACLs)

ACLs are sequential lists of `permit` or `deny` conditions that apply to network addresses or upper-layer protocols. They are used to filter traffic and restrict management access.

- **Implicit Deny:** Every ACL, regardless of type, has an **implicit** `deny any` at the end, meaning if traffic doesn't match an explicit `permit` statement, it is denied.

Configuration (Standard ACL)

- **What it is:** Filters traffic based **only on the source IP address**. Must be placed as close to the **destination** as possible.
- **Number Range:** 1-99 and 1300-1999.
- **Goal:** Deny traffic from host `192.168.10.5` but allow all other traffic into the `20.0.0.0/8` network.

1. **Create the ACL:** (Order matters)

   Cisco CLI

   ```
   Router(config)#access-list 1 deny host 192.168.10.5
   Router(config)#access-list 1 permit any
   ```

2. **Apply the ACL to the Interface:** (Inbound filter on the destination interface)

   Cisco CLI

   ```
   Router(config)#interface GigabitEthernet0/1
   Router(config-if)#ip access-group 1 in
   Router(config-if)#end
   ```

Configuration (Extended ACL)

- **What it is:** Filters traffic based on **source IP, destination IP, protocol (TCP/UDP/ICMP), and port number**. Must be placed as close to the **source** as possible.
- **Number Range:** 100-199 and 2000-2699.
- **Goal:** Deny host `192.168.1.10` from using Telnet (TCP port 23) to server `172.16.1.5`, but allow all other traffic.

1. **Create the ACL:** (Specify protocol and port)

   Cisco CLI

   ```
   Router(config)#access-list 101 deny tcp host 192.168.1.10 host 172.16.1.5
   eq 23
   Router(config)#access-list 101 permit ip any any
   ```

2. **Apply the ACL to the Interface:** (Outbound filter on the source interface)

   Cisco CLI

   ```
   Router(config)#interface GigabitEthernet0/0
   Router(config-if)#ip access-group 101 out
   Router(config-if)#end
   ```

Verification (ACL)

Cisco CLI

```
Router#show access-lists 101       ! View the ACL content and hit count
Router#show ip interface GigabitEthernet0/0 ! Verify ACL application
```

5.7 Configure and verify Layer 2 security features

These features protect the switch fabric from common attacks like DHCP starvation and ARP poisoning.

DHCP Snooping

- **What it is:** A Layer 2 security feature that acts as a firewall between untrusted hosts and trusted DHCP servers.
- **Function:** Maintains a binding database of valid IP-to-MAC address assignments. **Blocks malicious DHCP traffic** from unauthorized (untrusted) ports.
- **Goal:** Prevent rogue DHCP servers and DHCP starvation attacks.

Configuration (DHCP Snooping)

1. **Enable DHCP Snooping globally and specify VLANs:**

   Cisco CLI

   ```
   Switch(config)#ip dhcp snooping
   Switch(config)#ip dhcp snooping vlan 10,20
   ```

2. **Configure the interface connected to the trusted DHCP server as `trusted`:**

   Cisco CLI

   ```
   Switch(config)#interface GigabitEthernet0/1
   Switch(config-if)#ip dhcp snooping trust
   ```

Dynamic ARP Inspection (DAI)

- **What it is:** A Layer 2 security feature that prevents **ARP poisoning/spoofing** by validating ARP packets.
- **Function:** Compares the ARP packet (source IP/MAC) against the **DHCP Snooping binding database**. Invalid packets are dropped.
- **Requirement:** DHCP Snooping must be enabled first.

Configuration (DAI)

1. **Enable DAI for specified VLANs:**

   Cisco CLI

   ```
   Switch(config)#ip arp inspection vlan 10,20
   Switch(config)#end
   ```

Port Security

- **What it is:** A feature that restricts the number of valid MAC addresses allowed on a switch port.

- **Function:** Limits a port to a specified number of MAC addresses to prevent attackers from using unauthorized devices or performing MAC flooding attacks.
- **Violation Modes (Most Common):**
  - `shutdown` **(Default):** Shuts down the port and places it in an **err-disabled** state.
  - `restrict`**:** Drops packets from unauthorized MACs and sends an SNMP trap/syslog message.

Configuration (Port Security)

- **Goal:** Limit port F0/1 to one dynamically learned MAC address and shut down the port upon violation.

1. **Enable the port and set it to Access mode:**

   Cisco CLI

   ```
   Switch(config)#interface FastEthernet0/1
   Switch(config-if)#switchport mode access
   ```

2. **Enable port security:**

   Cisco CLI

   ```
   Switch(config-if)#switchport port-security
   ```

3. **Set the maximum number of allowed MAC addresses:**

   Cisco CLI

   ```
   Switch(config-if)#switchport port-security maximum 1
   ```

4. **Configure the MAC address to be learned dynamically and sticky:**

   Cisco CLI

   ```
   Switch(config-if)#switchport port-security mac-address sticky
   Switch(config-if)#end
   ```

Verification (L2 Security)

Cisco CLI

```
Switch#show ip dhcp snooping binding
Switch#show ip arp inspection
Switch#show port-security interface FastEthernet0/1
```

5.8 Compare authentication, authorization, and accounting concepts (AAA)

AAA is the framework for controlling access to network resources and recording user activity. It is typically implemented using a dedicated server like Cisco Identity Services Engine (ISE) or a RADIUS/TACACS+ server.

Concept	What it is	Function	FHRP Analogy
Authentication	**Who** are you? (Proof of Identity)	Verifying the identity of a user or device (e.g., username/password, certificate).	Showing your ID to the bouncer.
Authorization	**What** are you allowed to do? (Access Rights)	Granting specific permissions based on the verified identity (e.g., read-only access, privilege level 15).	The bouncer granting access to the VIP lounge.
Accounting	**What** did you do? (Audit Trail)	Recording the session details (start/stop time, commands executed, data transferred).	The security guard logging your entry time and activities.

Export to Sheets

Configuration (Basic AAA for Console Access)

- **Goal:** Use a RADIUS server (e.g., `10.1.1.20`) for authentication on VTY lines.

1. **Configure the server group:**

   Cisco CLI

```
Router(config)#aaa new-model
Router(config)#radius server RADIUS_SERVER
Router(config-radius-srv)#address ipv4 10.1.1.20 auth-port 1812 acct-port
1813
Router(config-radius-srv)#key sharedsecretkey
Router(config-radius-srv)#exit
```

2. **Apply AAA to VTY lines:**

   Cisco CLI

```
Router(config)#line vty 0 4
Router(config-line)#login authentication default
Router(config-line)#end
```

5.9 Describe wireless security protocols (WPA, WPA2, and WPA3)

Protocol	Standard	Encryption Cipher	Authentication	Security Level
WPA	WPA (Wi-Fi Protected Access)	TKIP (Temporal Key Integrity Protocol)	PSK or 802.1X	**Obsolete**. Vulnerable to attacks.
WPA2	IEEE 802.11i	**AES-CCMP** (Advanced	PSK or 802.1X	**Current Standard** for most

Protocol	Standard	Encryption Cipher	Authentication	Security Level
		Encryption Standard)		deployments. Highly secure.
WPA3	WPA3 (Wi-Fi Protected Access 3)	**AES-GCMP** (Stronger AES Mode)	SAE (Simultaneous Authentication of Equals) or 802.1X	**Newest Standard**. Mandatory use of AES. Provides better key exchange.
WEP	N/A (Original)	RC4	Static Key	**Extremely Vulnerable**. Never use.

Export to Sheets

5.10 Configure and verify WLAN within the GUI using WPA2 PSK

This process is typically done on a Wireless LAN Controller (WLC) or a cloud management platform (e.g., Cisco Meraki).

Step-by-Step Configuration (Conceptual - WLC GUI)

1. **Navigate to WLANs:** Open the WLC or Cloud Dashboard GUI and go to the "WLANs" or "SSIDs" section.
2. **Create New WLAN:** Click "Add New" and assign:
   - **Profile Name:** Internal name for the WLC (e.g., `Staff-VLAN10`).
   - **SSID:** Broadcast name for the clients (e.g., `Corp-WiFi`).
   - **VLAN ID:** Assign the WLAN to a specific VLAN (e.g., VLAN 10).
3. **Configure Security:** Navigate to the "Security" tab.
   - **Layer 2 Security:** Select **WPA2**.
   - **Authentication Key Management:** Select **PSK** (Pre-Shared Key).
   - **Cipher Suite:** Ensure **AES** is selected (CCMP).
4. **Set the PSK:** Enter the passphrase (e.g., `securepassword123`) that clients will use to connect.
5. **Apply and Verify:** Apply the changes. The APs will begin broadcasting the new SSID.

Verification (WLC)

- **WLC CLI:** `show wlan summary` (Verify the WLAN is enabled and mapped to the correct VLAN).
- **Client Side:** Attempt to connect a laptop/phone and verify it successfully obtains an IP address in the correct subnet (VLAN).

Domain 6.0 Automation and Programmability

6.1 Explain how automation impacts network management

Network automation fundamentally changes how networks are managed, moving from manual, device-by-device configuration to programmatic, centralized control.

Impact Area	Traditional (Manual) Approach	Automated Approach
Speed & Agility	Slow, serial changes; difficult to scale or respond quickly to business needs.	Fast, simultaneous changes across thousands of devices; rapid deployment of new services.
Consistency	High risk of human error (misconfigurations, typos); configuration drift between devices.	Guaranteed **consistency** and adherence to golden templates; eliminates human error.
Efficiency	Repetitive, tedious tasks (e.g., configuring 50 VLANs).	Engineers focus on design and strategy; tasks are executed by scripts/tools.
Cost	High operational expenditure (OpEx) due to labor hours and troubleshooting human errors.	Reduced OpEx due to efficiency and fewer outage-causing errors.
Scale	Difficult and risky to manage large-scale changes.	Easy to execute large-scale changes simultaneously and safely.

6.2 Compare traditional networks with controller-based networking

Feature	Traditional Network (CLI-based)	Controller-Based Network (SDN)
Intelligence	Distributed (Each device makes its own decisions).	**Centralized** (A single controller makes global decisions).
Configuration	Manual configuration on each device (CLI, SNMP).	Programmatic configuration via the **Controller's API**.
Protocols Used	Traditional protocols (OSPF, STP, HSRP, Telnet/SSH).	**APIs** (REST), **Encapsulation protocols** (VXLAN, NVGRE).
Deployment	Slow, device-by-device, complex troubleshooting.	Fast, template-driven, simplified policy deployment.
Management	Individual device logging; troubleshooting requires device hop-by-hop investigation.	Centralized visibility and control from a single dashboard.

6.3 Describe controller-based, software defined architecture

Controller-based architecture, also known as **Software-Defined Networking (SDN)**, separates the network logic from the hardware forwarding plane.

6.3.a Separation of control plane and data plane

- **Data Plane (Forwarding Plane):**
  - **Function:** Handles the **actual forwarding** of traffic (packets and frames).
  - **Location:** Resides on the network device (e.g., router ASIC, switch ASIC).
  - **Action:** Uses information provided by the control plane (e.g., forwarding table/FIB) to make fast, per-packet decisions.
- **Control Plane:**
  - **Function:** Builds the **network topology map**, runs routing protocols, and creates the forwarding tables.
  - **Location (SDN):** Resides in the **centralized controller**.
  - **Action:** Directs the data plane how and where to forward traffic based on global network policy.

6.3.b Northbound and Southbound APIs

- **Southbound API (Controller to Network Devices):**
  - **Function:** The interface used by the **Controller** to communicate with and program the underlying **network devices**.
  - **Purpose:** Allows the centralized controller to push configuration, policies, and forwarding rules (control plane logic) down to the data plane devices.
  - **Examples:** NETCONF, RESTCONF, OpenFlow.
- **Northbound API (Controller to Applications/Users):**
  - **Function:** The interface used by **Applications** (e.g., orchestration tools, custom scripts, monitoring systems) to communicate with the **Controller**.
  - **Purpose:** Allows administrators and applications to easily request network services or push high-level policy changes without needing to know device-level commands.
  - **Examples:** REST API (HTTP/HTTPS calls).

6.4 Explain AI (generative and predictive) and machine learning in network operations

- **Machine Learning (ML) in Networking:**
  - **What it is:** Algorithms that allow systems to learn patterns and make decisions based on large amounts of network data without being explicitly programmed for every scenario.
  - **Use Case: Predictive Analytics.** Identifying abnormal traffic patterns that indicate a security threat or anticipating network congestion before it occurs by analyzing historical utilization trends.
- **AI (Artificial Intelligence) in Networking:**

- **What it is:** The umbrella field encompassing ML, enabling systems to perform cognitive functions.
- **Predictive AI:** Uses ML to forecast future outcomes (e.g., predicting when a hard drive on a server will fail or predicting the need for a bandwidth upgrade).
- **Generative AI:** Focuses on creating new content.
- **Use Case: Troubleshooting and Configuration:** Generating natural language descriptions of network issues or automatically generating configuration templates based on high-level design requirements.

6.5 Describe characteristics of REST-based APIs

Representational State Transfer (REST) is an architectural style used to define web services. **REST APIs** are the dominant standard for modern northbound and southbound communication.

Characteristic	Description
Authentication Types	Methods used to verify the user or application making the API request: **Basic Authentication** (username/password, often base64 encoded) and **Token-based Authentication** (client gets a temporary access token after initial login).
CRUD	The four fundamental operations an API can perform on a resource: **C**reate, **R**ead, **U**pdate, and **D**elete.
HTTP Verbs (Methods)	The specific actions the client requests to be performed on the resource: **GET** (Read), **POST** (Create), **PUT** (Update/Replace), **PATCH** (Update/Modify), and **DELETE** (Delete).
Data Encoding	The format used to send and receive data between the client and the API server. The most common format is **JSON** (JavaScript Object Notation), with **XML** also supported.

6.6 Recognize the capabilities of configuration management mechanisms

Configuration management tools automate the deployment, configuration, and monitoring of large groups of network devices or servers, replacing manual device-by-device configuration.

- **Ansible:**
  - **Type:** Agentless configuration management and orchestration tool.
  - **Language:** Uses YAML for human-readable **Playbooks** (instructions).
  - **Capability:** Pushes configuration changes to network devices using SSH and relies on device CLI/APIs. Excellent for simpler tasks and rapid deployment due to its agentless nature.
- **Terraform:**
  - **Type:** Infrastructure as Code (IaC) tool focused on provisioning and managing infrastructure lifecycle.

- **Language:** Uses **HCL** (HashiCorp Configuration Language).
- **Capability:** Manages the life cycle of infrastructure components (**Create, Read, Update, Delete**) in cloud platforms (AWS, Azure) and network devices. Ideal for standing up or tearing down entire network environments and ensuring the end state of the network matches the declared code.

6.7 Recognize components of JSON-encoded data

JSON (JavaScript Object Notation) is a lightweight, human-readable data format used extensively in REST APIs for data exchange.

- **Syntax:** Data is represented as **key-value pairs** separated by commas.
- **Key-Value Pair:** The fundamental unit of JSON data, where the **Key** is always a string (in double quotes) and the **Value** can be one of several data types.

Components:

1. **Object (Key-Value Pairs):**
   - Starts with an opening curly brace `{` and ends with a closing curly brace `}`.
   - Represents an entity or dictionary.
   - **Example:** `{"hostname": "RTR-CORE-01"}`
2. **Array (Ordered List of Values):**
   - Starts with an opening square bracket `[` and ends with a closing square bracket `]`.
   - Values are separated by commas.
   - **Example:** `["Gi1/0/1", "Gi1/0/2", "Gi1/0/3"]`
3. **Values:** Can be one of the following data types:
   - **String:** Text enclosed in double quotes (e.g., `"Cisco IOS XE"`).
   - **Number:** Integer or floating point (e.g., `100` or `1.5`).
   - **Boolean:** `true` or `false`.
   - **`null`:** Represents no value.
   - **Object:** A nested JSON object.
   - **Array:** A nested JSON array.

Example of JSON-Encoded Data:

JSON

```
{
  "device": "SW-ACCESS-01",
  "interfaces": [
    {
```

```
      "name": "GigabitEthernet1/0/1",
      "vlan_id": 10,
      "status": true
    },
    {
      "name": "GigabitEthernet1/0/2",
      "vlan_id": 20,
      "status": false
    }
  ],
  "location": "Floor 5"
}
```

20 CCNA Preparation Exercises

1. Network Fundamentals & IP Addressing

#	Task Description	Expected Outcome
1.	**Subnetting & Host Configuration:** Given the network `192.168.50.0 /26`, configure three hosts (PC-A, PC-B, PC-C) with the first three usable IP addresses in that subnet, using the correct subnet mask and a gateway of `.62`.	PC-A, PC-B, and PC-C can successfully ping the gateway IP address (`192.168.50.62`).
2.	**IPv6 Link-Local:** Configure a Cisco router's interface G0/0/0 with a Global Unicast Address (`2001:db8:acad:1::1/64`) and verify the automatically generated Link-Local address.	The output of `show ipv6 interface G0/0/0` displays both the GUA and a Link-Local address starting with `FE80`.
3.	**TCP vs. UDP Service Check:** Use `netstat` on a host device to determine if the DNS server is using TCP or UDP for a standard query, and document the port.	State whether DNS is primarily using **TCP** or **UDP** and the corresponding **port number** observed.
4.	**L2 Troubleshooting:** Connect two switches via a copper cable but intentionally configure one side for `full-duplex` and the other for `half-duplex`.	The output of `show interfaces` on the full-duplex switch clearly shows a high count of **CRC errors** or **late collisions**.

2. Network Access (Switching)

#	Task Description	Expected Outcome
5.	**VLANs & Trunks:** Configure two switches (SW1, SW2) with VLANs 10 (Data) and 20 (Voice). Configure the link between them as an 802.1Q trunk.	A host in VLAN 10 connected to SW1 can successfully ping a host in VLAN 10 connected to SW2. The output of `show interfaces trunk` shows VLANs 10 and 20 passing.

#	Task Description	Expected Outcome
6.	**InterVLAN Routing (SVI):** Using a Layer 3 switch, configure SVIs for VLAN 10 and VLAN 20, assigning them gateways `192.168.10.1` and `192.168.20.1` respectively.	A host in VLAN 10 can successfully ping a host in VLAN 20, and `show ip route` on the switch shows two **Connected** routes.
7.	**EtherChannel with LACP:** Aggregate two physical links between SW1 and SW2 into a logical port channel using **LACP active** mode.	The output of `show etherchannel summary` confirms the Port-Channel (Po) is up (status **SU**) and using LACP (**LACP**).
8.	**STP Root Bridge Control:** Force a specific switch (SW-A) to become the **Root Bridge** for VLAN 10 using the `priority` command.	The output of `show spanning-tree vlan 10` on SW-A displays "This bridge is the root" for VLAN 10.
9.	**Port Security:** Configure a switch access port (F0/1) to allow only the **first learned MAC address** and set the violation mode to **shutdown**.	When a second, different device is connected to F0/1, the port immediately transitions to the **err-disabled** state.
10.	**LLDP/CDP Verification:** Connect a Cisco router and a non-Cisco (or simulated non-Cisco) device to a switch. Use both discovery protocols to gather neighbor information.	The router's model and connected interface are displayed in the `show cdp neighbors` output, and the non-Cisco device is visible only in the `show lldp neighbors` output.

3. IP Connectivity (Routing)

#	Task Description	Expected Outcome
11.	**IPv4 Static Routing:** Configure two routers (R1, R2) with a non-directly connected network (`172.16.1.0/24`) behind R2 using a **next-hop IP address** static route on R1.	R1 can successfully ping a host address in the `172.16.1.0/24` network.
12.	**Floating Static Route:** Configure a primary dynamic route (e.g., OSPF) to R2's network, and a backup **floating static route** via R3 with an Administrative Distance (AD) of 150.	The floating static route is visible only in the routing table when the **primary OSPF route fails** (e.g., R2's interface is shut down).
13.	**Single Area OSPFv2:** Configure three routers (R1, R2, R3) in a single OSPF Area 0, ensuring they all become neighbors and advertise their connected networks.	`show ip ospf neighbor` on R1 shows R2 and R3 in the **FULL** state, and R1's routing table contains all networks from R2 and R3 with a code of **O**.
14.	**HSRP Configuration:** Configure two routers (R1, R2) to run HSRP, setting R1 as the **Active** router and R2 as **Standby** for group 1, using a VIP of `192.168.1.254`.	The output of `show standby brief` on both routers confirms R1 is in the **Active** state and R2 is in the **Standby** state.

4. IP Services

#	Task Description	Expected Outcome
15.	**Inside Source PAT (Overload):** Configure a router (R1) to translate all private internal traffic (`10.0.0.0/8`) to its **outside interface's public IP address** using overload (PAT).	An internal host can access an outside web server, and `show ip nat translations` on R1 displays the host's private IP, mapped to the public IP using a **unique port number**.
16.	**DHCP Relay:** Configure a router (R1) to act as a **DHCP Relay Agent** for clients on its G0/0/0 interface, forwarding requests to a DHCP server at `172.16.2.5`.	A client connected to R1's G0/0/0 receives an IP configuration from the DHCP server `172.16.2.5`.
17.	**Secure Remote Access:** Configure a router for remote management using **SSH** (modulus 1024) and disable the unsecure Telnet protocol on the VTY lines.	A remote PC can successfully SSH to the router, and attempting a Telnet connection results in a **connection refused/denied** message.

5. Security Fundamentals

#	Task Description	Expected Outcome
18.	**Extended ACL Filtering:** Create and apply an extended ACL that **denies** only HTTP traffic (TCP port 80) from a specific source subnet (`192.168.5.0/24`) to a specific destination subnet (`192.168.10.0/24`), while permitting all other traffic.	Hosts in the `192.168.5.0/24` subnet can **ping** hosts in the destination subnet, but **cannot access** a web server running on TCP port 80 in that destination subnet.

6. Automation and Programmability

#	Task Description	Expected Outcome
19.	**JSON Data Interpretation:** Given a JSON-encoded data snippet detailing router interfaces, identify the data type (string, array, or number) of the **Interface name** field and the **Bandwidth** field.	State the data type for **Interface name** (e.g., string) and the data type for **Bandwidth** (e.g., number).
20.	**REST API Concept Mapping:** Map the following HTTP verbs to the correct CRUD operation: **GET, POST, DELETE**.	Successfully match **GET** to its CRUD operation, **POST** to its CRUD operation, and **DELETE** to its CRUD operation.

5 Advanced CCNA Capstone Exercises

#	Task Description	Domain Focus	Expected Outcome
21.	**Multi-Area OSPF & LSA Verification:** Configure three routers (R1, R2, R3) in an OSPF topology with R2 as the **Area**	**Routing (L3)**	R1's OSPF database shows **Type 1** and **Type 3** LSAs (but no Type 2 from Area 1), and R1 successfully pings a

#	Task Description	Domain Focus	Expected Outcome
	Border Router (ABR) connecting **Area 0** (R1-R2 link) and **Area 1** (R2-R3 link).		loopback address on R3, verifying inter-area routing.
22.	**Secure InterVLAN using RoaS and Extended ACLs:** Configure **Router-on-a-Stick (RoaS)** on R1 to route between VLAN 10 (IT) and VLAN 20 (Guest). Apply an **Extended ACL** on the router's sub-interface to explicitly deny all ICMP traffic between VLAN 10 and VLAN 20, while permitting all other IP traffic.	**Switching (L2) & Security (L5)**	A host in VLAN 10 can **Telnet** (or SSH) to a host in VLAN 20, but a **ping** attempt between the two VLANs fails and increments the `deny` hit count on the applied ACL.
23.	**DHCP Snooping with ARP Inspection:** Enable DHCP Snooping on a switch for VLAN 10. Configure the uplink port to the DHCP server as **trusted** and enable **Dynamic ARP Inspection (DAI)** on the same VLAN. Statically bind a host MAC/IP address to an untrusted port.	**Security (L5)**	The output of `show ip dhcp snooping binding` shows the static MAC/IP entry. An attempt to **ARP spoof** the gateway from another device on an untrusted port is **blocked** by the switch, indicated by a DAI log message.
24.	**AAA with Tacacs+ Configuration:** Configure a router to use a **TACACS+** server (simulated or real) for **Authentication** and **Authorization** of management access (VTY lines). Define a local fallback user named `breakglass` with privilege level 15.	**Security (L5)**	A user logging in via VTY is authenticated by the TACACS+ server. When the TACACS+ server is made **unreachable** (simulated server failure), the `breakglass` user can successfully log in using the **local** database.
25.	**WPA3-Enterprise Concept Review:** Describe the core difference between the key exchange mechanisms in **WPA2-PSK** and **WPA3-Personal** (SAE), and the required network element (beyond the AP) for **WPA3-Enterprise**.	**Wireless (L5) & Fundamentals (L1)**	Identify **SAE** as the WPA3-Personal mechanism and explain how it prevents **offline dictionary attacks**. Clearly state that a **RADIUS server** is the mandatory external component for WPA3-Enterprise.

🧠 CCNA Subnetting Memorization Table

This table is built around the **power of 2** and the **number of bits borrowed/available**.

Prefix (CIDR)	Subnet Mask (Netmask)	Block Size / Increment	Usable Host IPs (2h−2)
/30	255.255.255.**252**	**4**	**2**

Prefix (CIDR)	Subnet Mask (Netmask)	Block Size / Increment	Usable Host IPs (2h−2)
/29	255.255.255.**248**	**8**	**6**
/28	255.255.255.**240**	**16**	**14**
/27	255.255.255.**224**	**32**	**30**
/26	255.255.255.**192**	**64**	**62**
/25	255.255.255.**128**	**128**	**126**
---	**---**	**---**	**---**
/24	255.255.255.0	256	254
/23	255.255.254.0	512	510
/22	255.255.252.0	1,024	1,022
/16	255.255.0.0	65,536	65,534

Guide to Using the Table

1. The Key Column: Block Size (or Magic Number)

The **Block Size** is the most important value to memorize. It represents:

1. The **increment** between valid network addresses.
2. The value that the **subnet mask subtracts from 256**.

- *Example:* For a **/26**, the block size is **64**.
  - Subnets are: 0, 64, 128, 192.
  - Netmask is $256 - 64 = **192**$ (i.e., $255.255.255.192$).

2. Deriving Host Count

The **Usable Host IPs** is the total number of addresses in the block ($\mathbf{2}^{\mathbf{h}}$) minus the network address and the broadcast address ($-\mathbf{2}$).

- *Example:* For a **/29**, you have 3 host bits ($32 - 29 = 3$).
  - Total IPs: $2^3 = **8**$ (which is the Block Size).
  - Usable IPs: $8 - 2 = **6**$.

3. Deriving Subnet Mask (Last Octet Only)

To find the last octet of the subnet mask, memorize the relationship to the block size:

Subnet Mask	Block Size
.252	4
.248	8
.240	16
.224	32
.192	64
.128	128

4. Calculating Number of Networks (Subnets)

This is context-dependent, but generally, if you start from the default mask:

- **Number of Networks** $= \mathbf{2}^{\mathbf{s}}$, where 's' is the number of **subnet bits borrowed** (starting from the default class boundary).

CIDR (from /24)	Subnet Bits (s)	Number of Networks (2s)
/25	1	2
/26	2	4
/27	3	8
/28	4	16
/29	5	32
/30	6	64

- *Example:* Moving from a default **/24** to a **/27** means you borrowed **3 bits**. You can create $2^3 = \mathbf{8}$ subnets.